



CHAPTER

2

MEng2132

Engineering Thermodynamics I

Lecture Notes:

Faculty of Mechanical and Industrial Engineering,
Bahir Dar Institute of Technology, Bahir Dar, Ethiopia

Properties of Pure Substances



Objective of the chapter

- ❖ The objectives of this chapter are to introduce property relations relevant to engineering thermodynamics and provide several examples illustrating the use of the closed system energy balance together with the property relations considered in this chapter.

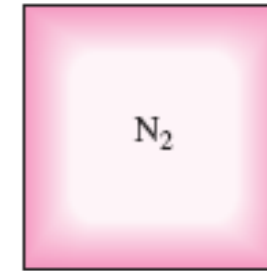


Learning outcomes

- When you complete your study of this chapter, you will be able to...
- ❖ demonstrate understanding of key concepts . . . including phase and pure substance,
- ❖ state principle for simple compressible systems, saturation temperature and saturation pressure, two-phase liquid-vapor mixture, quality, enthalpy, and specific heats.
- ❖ apply the closed system energy balance with property data.
- ❖ sketch T – y , p – y , and phase diagrams, and locate states on these diagrams.
- ❖ retrieve property data from Tables A-1 through A-23.
- ❖ apply the ideal gas model for thermodynamic analysis.

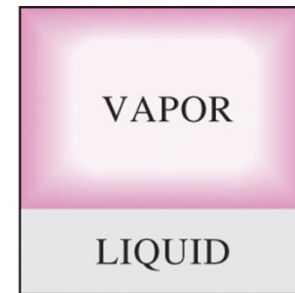
What is Pure Substances?

- ❖ A substance that has a fixed chemical composition throughout is called a **pure substance**.
- ❖ A pure substance does not have to be of a single chemical element or compound, however. A mixture of various chemical elements or compounds also qualifies as a pure substance as long as the mixture is *homogeneous*.
- ❖ Air, for example, is a mixture of several gases, but it is often considered to be a pure substance because it has a uniform chemical composition.
- ❖ However, a mixture of oil and water is not a pure substance. Since oil is not soluble in water, it will collect on top of the water, forming two chemically dissimilar regions.


 N_2


AIR

Nitrogen and gaseous air are pure substances.


 (a) H_2O


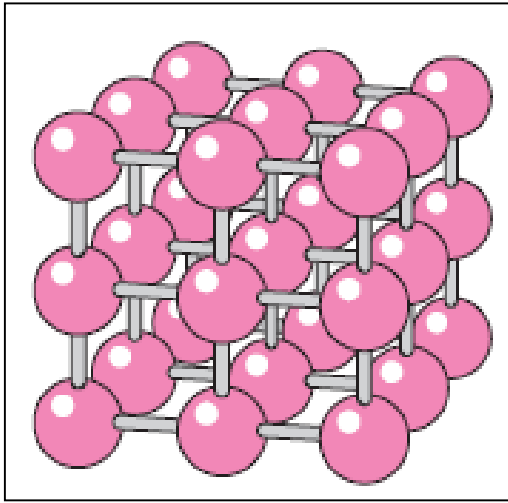
(b) AIR

A mixture of liquid and gaseous water is a pure substance, but a mixture of liquid and gaseous air is not.

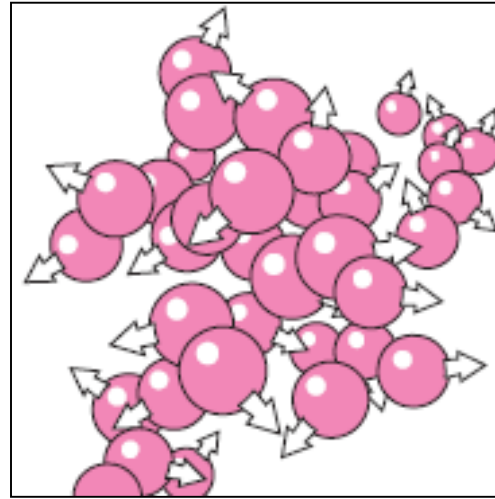


Phases of A Pure Substance

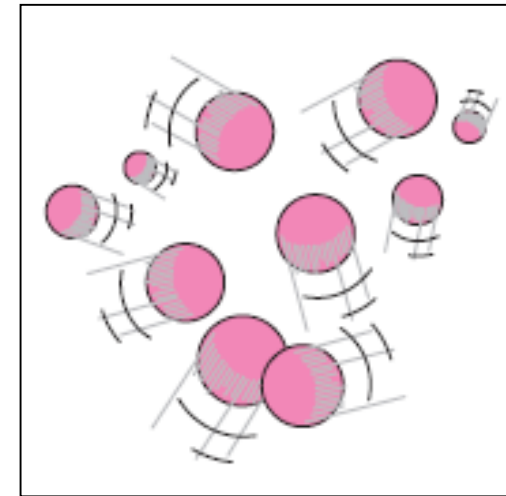
- ❖ The substances exist in different phases, e.g. at room temperature and pressure, copper is solid mercury, is a liquid and nitrogen is a gas.
- ❖ Under different conditions, each may appear in a different phase.
- ❖ Even though there are three principal phases—solid, liquid, and gas
- ❖ a substance may have several phases within a principal phase, each with a different molecular structure.
- ❖ Carbon, for example, may exist as graphite or diamond in the solid phase. Helium has two liquid phases; iron has three solid phases. Ice may exist at seven different phases at high pressures.

*Solid*

- ❖ **Solid: strong intermolecular bond**
- ❖ **molecules are closely packed together**
- ❖ **Because of the small distances between molecules in a solid, the attractive forces of molecules on each other are large and keep the molecules at fixed positions.**
- ❖ **In a solid, the attractive and repulsive forces between the molecules tend to maintain them at relatively constant distances from each other**

*Liquid*

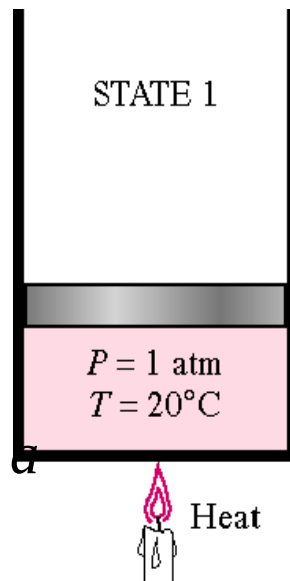
- ❖ **Liquid: intermediate intermolecular bonds**
- ❖ **The molecular spacing in the liquid phase is not much different from that of the solid phase, except the molecules are no longer at fixed positions relative to each other and they can rotate and translate freely.**

*Gas*

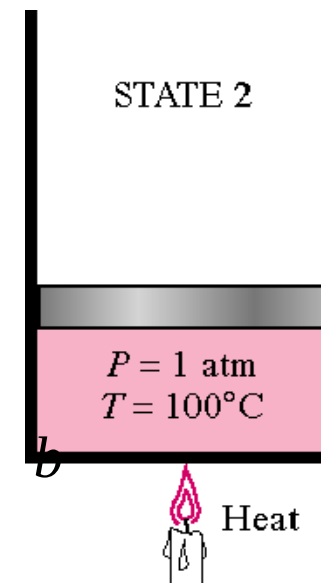
- ❖ **Gas: weak intermolecular bond**
- ❖ **they are separated by relatively large distances**
- ❖ **Molecules in the gas phase are at a considerably higher energy level than they are in the liquid or solid phases. Therefore, the gas must release a large amount of its energy before it can condense or freeze.**

Phase-change Processes of Pure Substances

- ❖ There are many practical situations where two phases of a pure substances coexist in equilibrium.
- ❖ E.g. water exists as a mixture of liquid and vapor in the boiler and etc.
- ❖ Consider a piston–cylinder device containing liquid water at 20°C and 1 atm
- ❖ pressure (state 1, Fig. a)

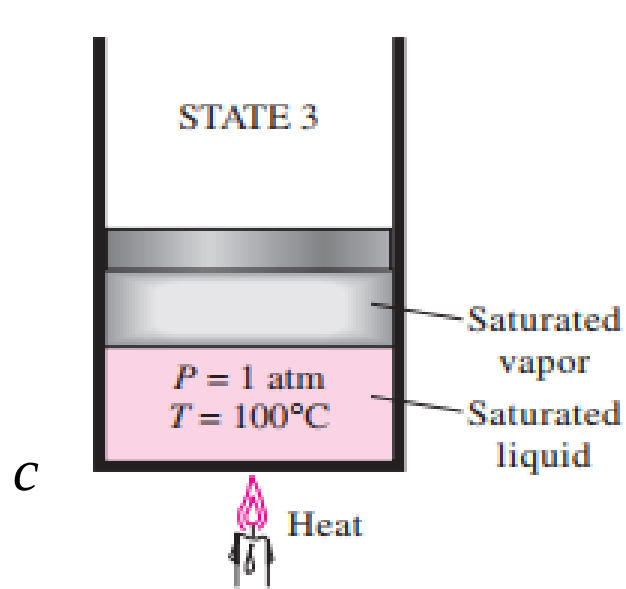


At 1 atm and 20°C, water exists in the liquid phase (*compressed liquid*).

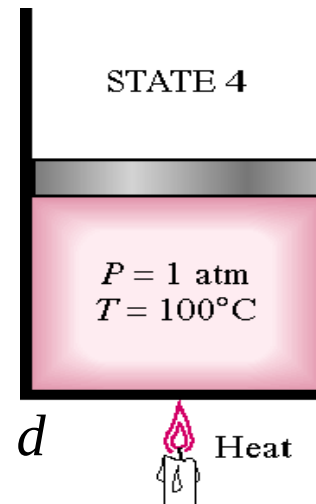


At 1 atm pressure and 100°C, water exists as a liquid that is ready to vaporize (*saturated liquid*).

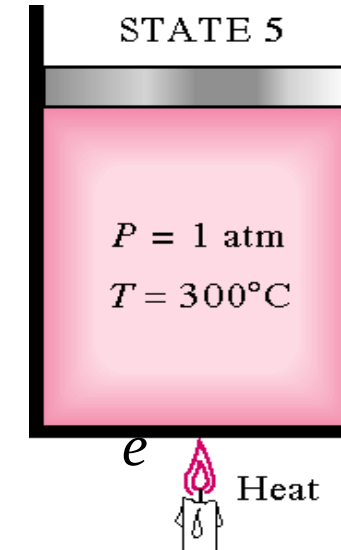
Phase-change Processes of Pure Substances



As more heat is transferred, part of the saturated liquid vaporizes (*saturated liquid–vapor mixture*).



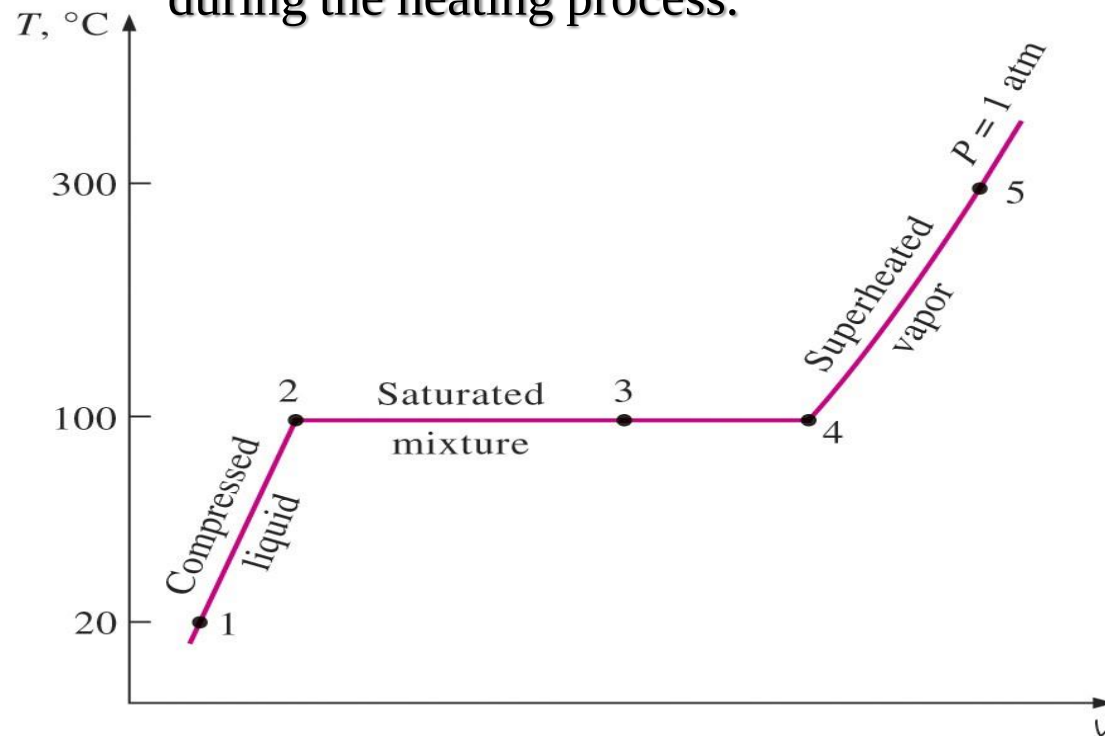
At 1 atm pressure, the temperature remains constant at 100°C until the last drop of liquid is vaporized (*saturated vapor*).



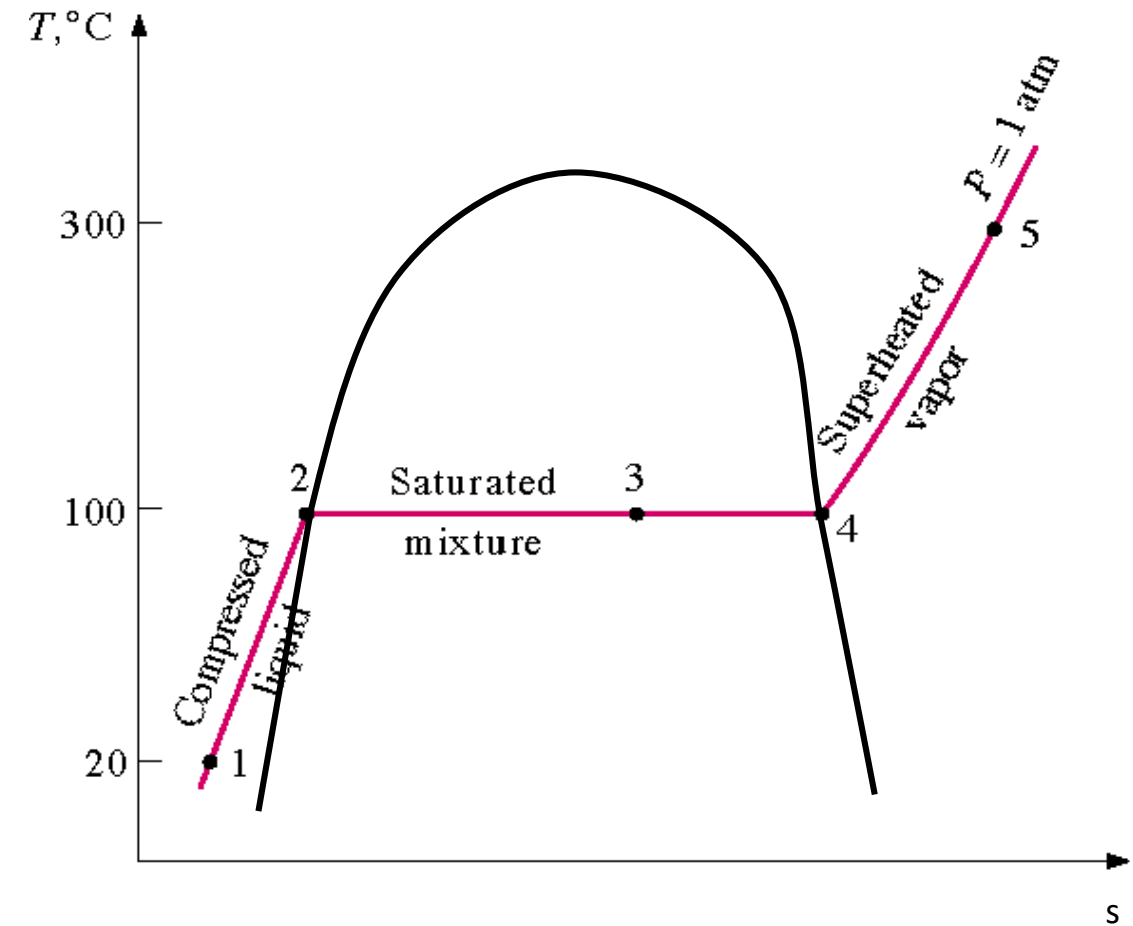
As more heat is transferred, the temperature of the vapor starts to rise (*superheated vapor*).

Phase-change Processes of Pure Substances

- ❖ If the entire process between state 1 and 5 described in the figure is reversed by cooling the water while maintaining the pressure at the same value, the water will go back to state 1, retracing the same path, and in so doing, the amount of heat released will exactly match the amount of heat added during the heating process.

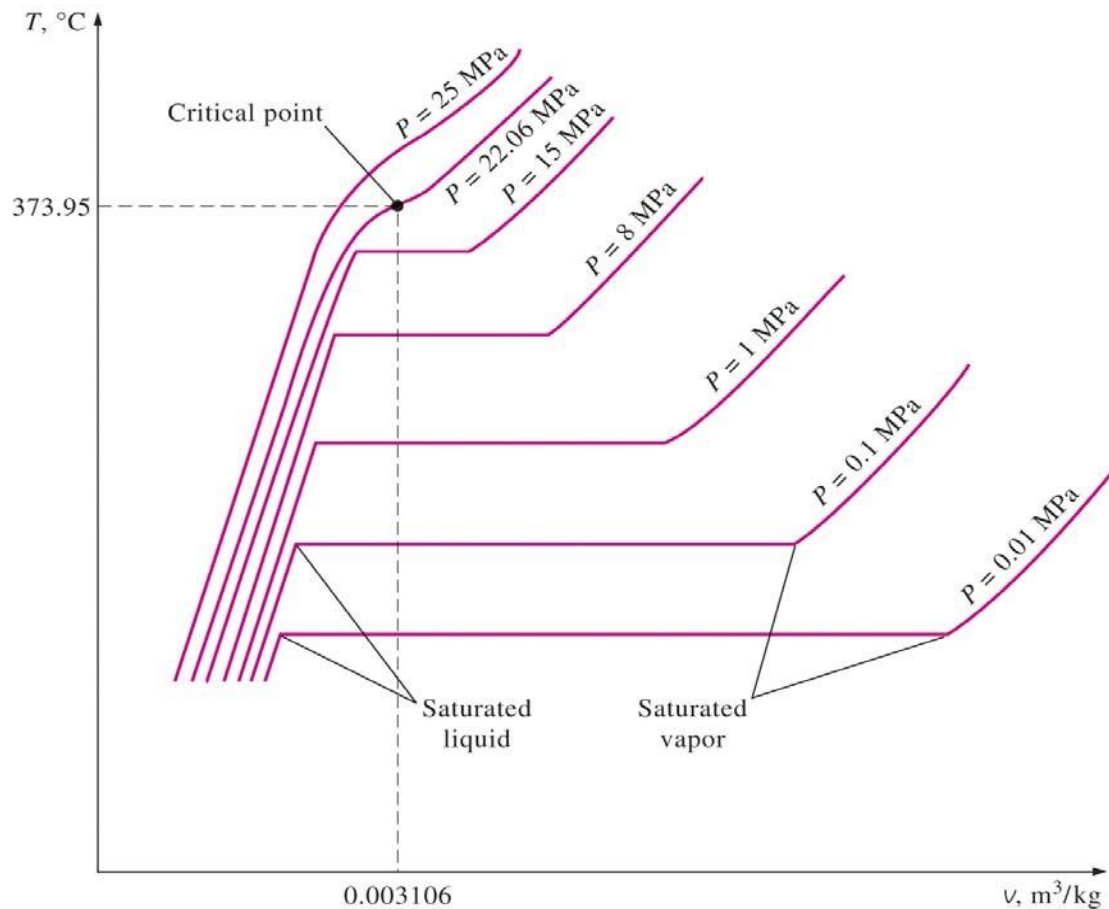


T - v diagram for the heating process of water at constant pressure.



PROPERTY DIAGRAMS FOR PHASE- CHANGE PROCESSES

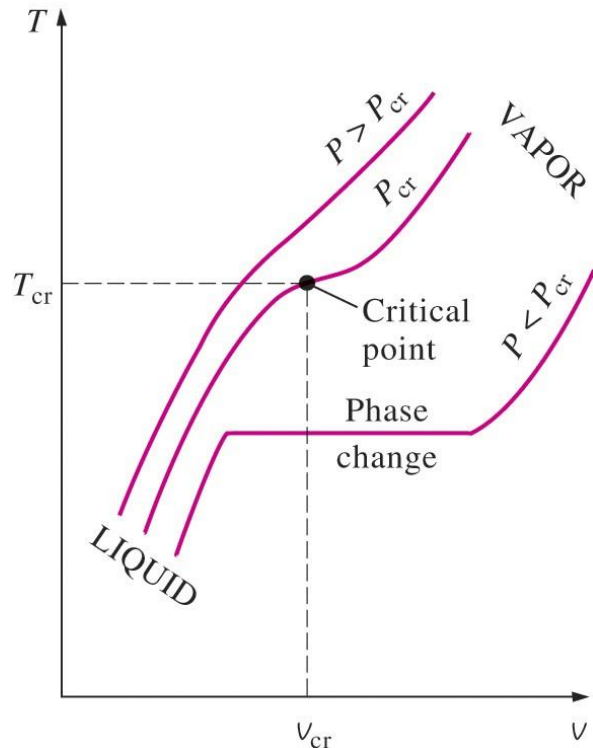
- The variations of properties during phase-change processes are best studied and understood with the help of property diagrams such as the T - v , P - v , and P - T diagrams for pure substances.



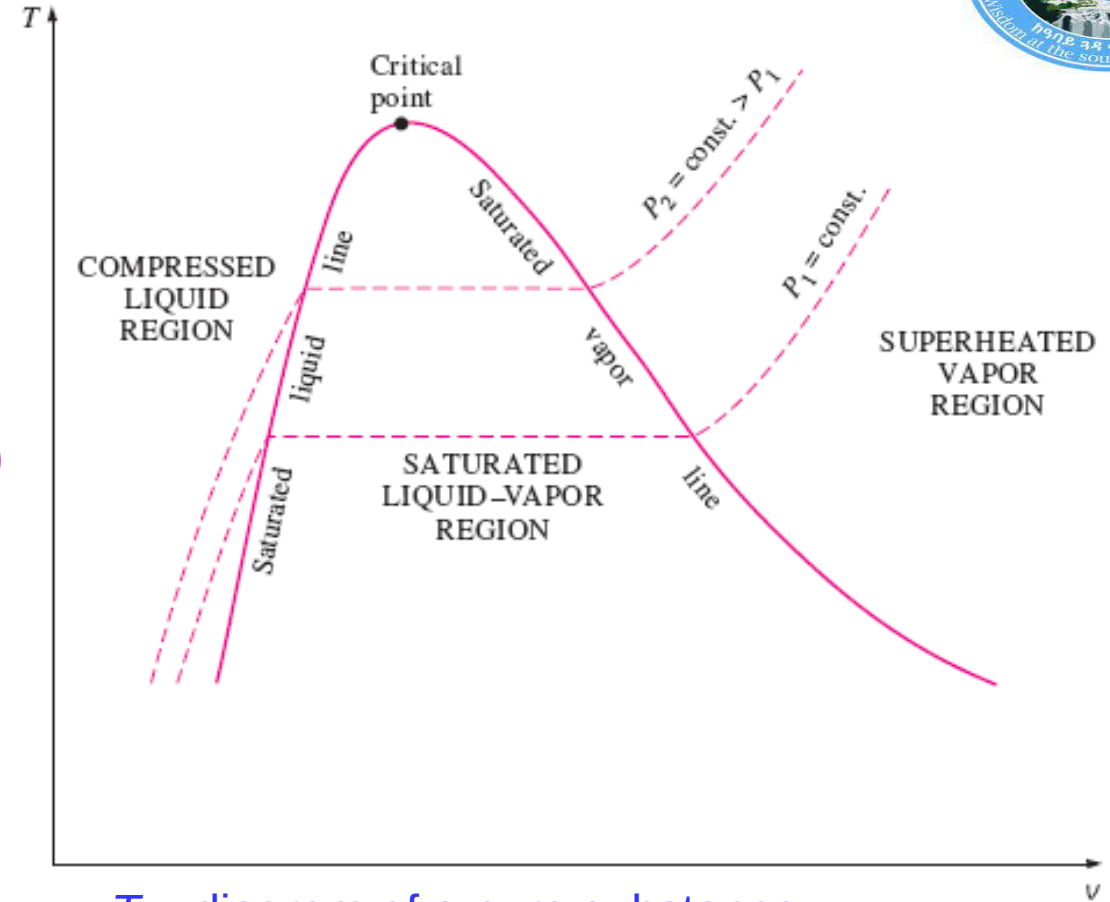
T - v diagram of constant-pressure phase-change processes of a pure substance at various pressures (numerical values are for water).

Property Diagram

- saturated liquid line
- saturated vapor line
- compressed liquid region
- superheated vapor region
- saturated liquid–vapor mixture region (wet region)



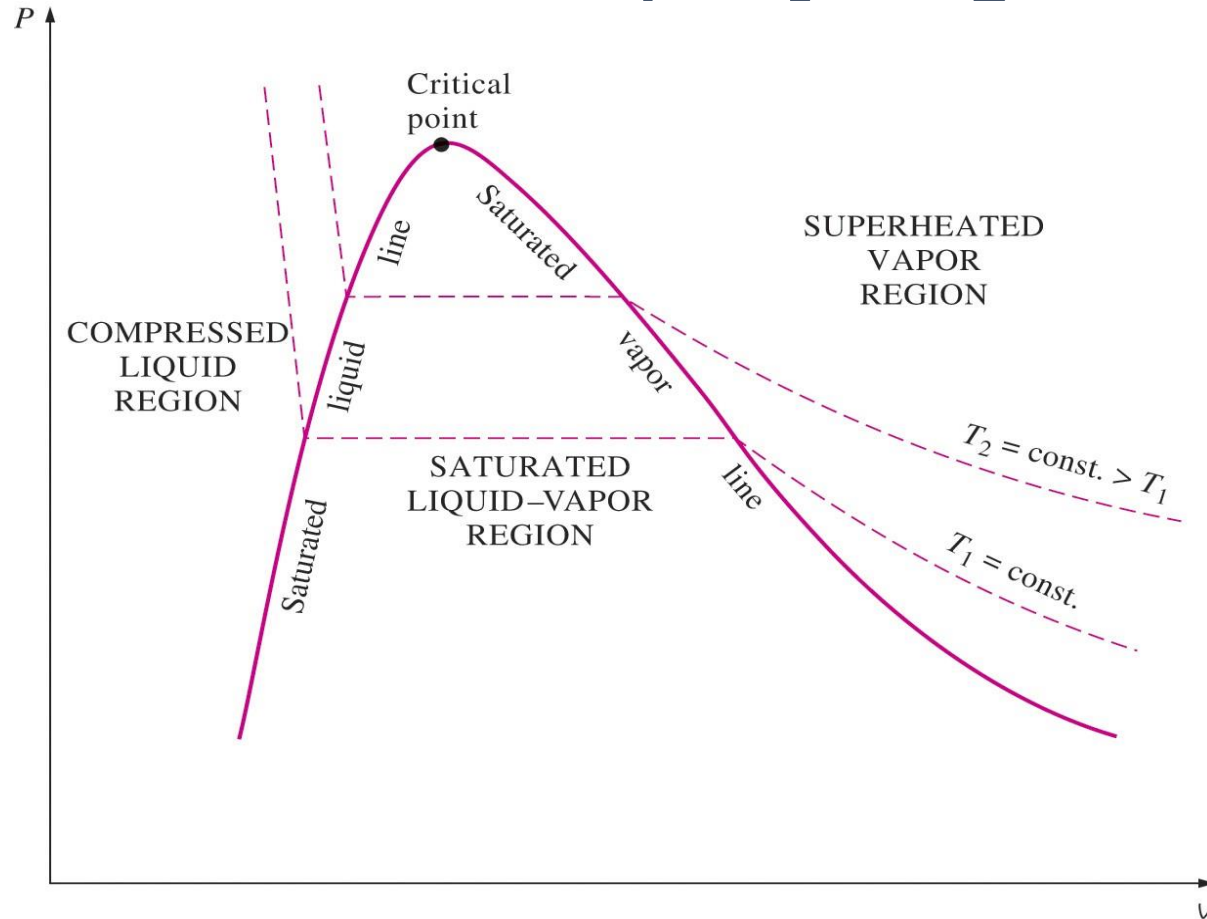
At supercritical pressures ($P > P_{cr}$), there is no distinct phase-change (boiling) process.



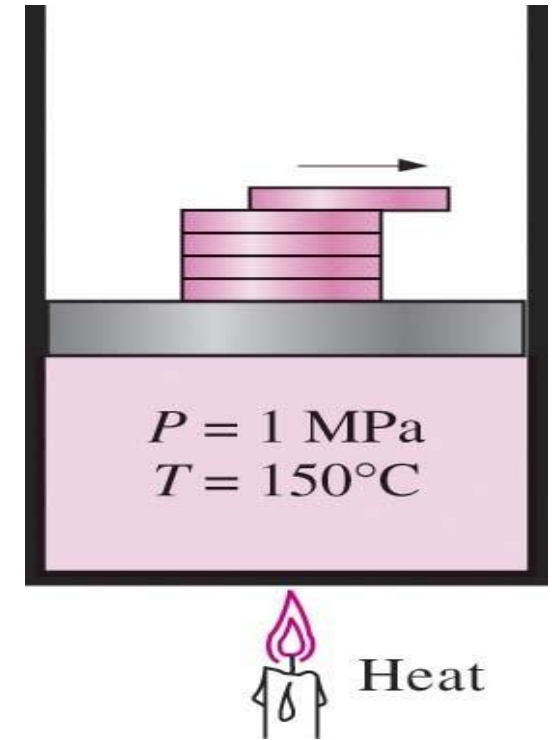
T-v diagram of a pure substance.

Critical point: The point at which the saturated liquid and saturated vapor states are identical.

Property Diagram



P - v diagram of a pure substance.

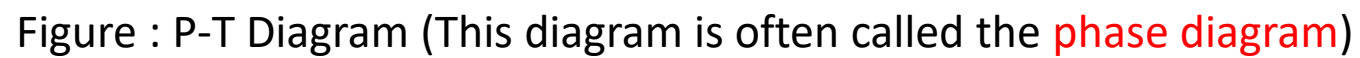


The pressure in a piston–cylinder device can be reduced by reducing the weight of the piston.



Important Definition

- **Critical point** - the temperature and pressure above which there is no distinction between the liquid and vapor phases.
- **Triple point** - the temperature and pressure at which all three phases can exist in equilibrium.
- **Sublimation** - change of phase from solid to vapor.
- **Vaporization** - change of phase from liquid to vapor.
- **Condensation** - change of phase from vapor to liquid.
- **Fusion or melting** - change of phase from solid to liquid.





Saturation

- ❖ *Saturation* is defined as a condition in which a mixture of vapor and liquid can exist together at a given temperature and pressure.
- ❖ *Saturation pressure* is the pressure at which the liquid and vapor phases are in equilibrium at a given temperature
- ❖ For a pure substance there is a definite relationship between saturation pressure and saturation temperature. The higher the pressure, the higher the saturation temperature

The graphical representation of this relationship between temperature and pressure at saturated conditions is called the *vapor pressure curve*

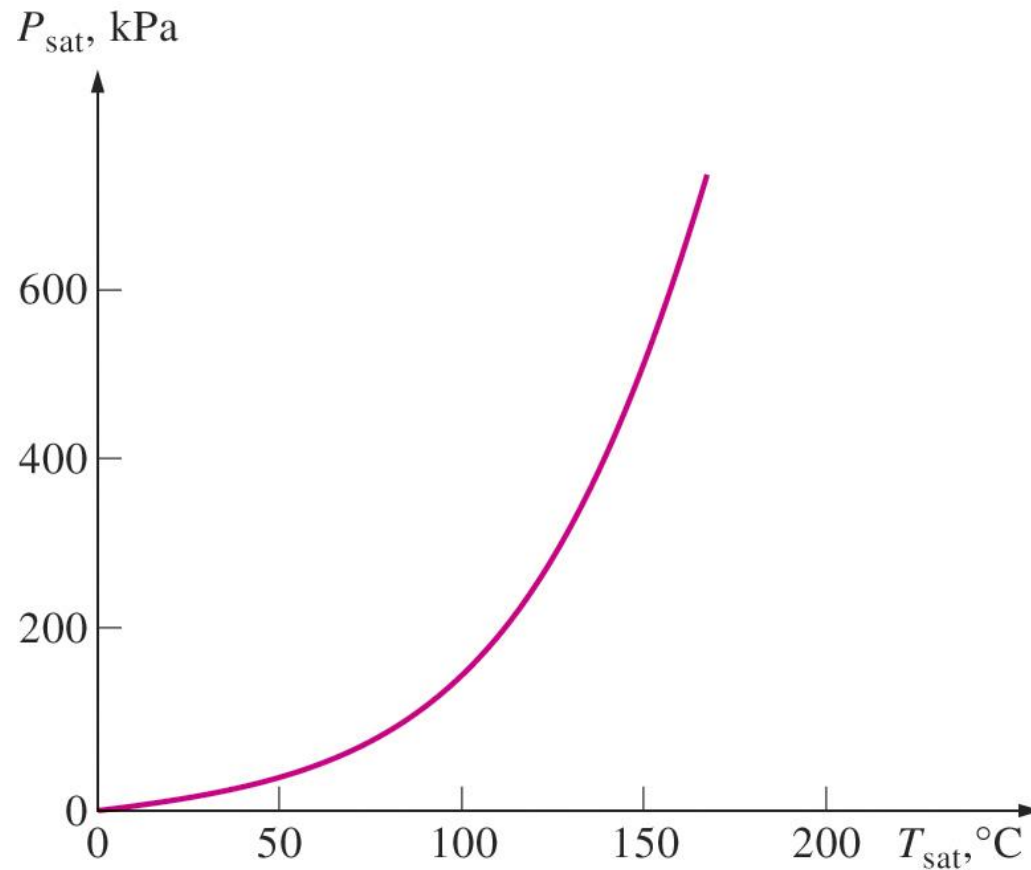


TABLE 3–1

Saturation (boiling) pressure of water at various temperatures

Temperature, T , °C	Saturation pressure, P_{sat} , kPa
–10	0.26
–5	0.40
0	0.61
5	0.87
10	1.23
15	1.71
20	2.34
25	3.17
30	4.25
40	7.39
50	12.35
100	101.4
150	476.2
200	1555
250	3976
300	8588

Saturated and Sub-cooled Liquids

- ❖ If a substance exists as a liquid at the saturation temperature and pressure, it is called a *saturated liquid*
- ❖ If the temperature of the liquid is lower than the saturation temperature for the existing pressure, it is called either a *subcooled liquid* or a *compressed liquid*

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Saturated and Superheated Vapors

TABLE 3–1

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100	101.4
150	476.2
200	1555
250	3976
300	8588

- ❖ If a substance exists entirely as vapor at saturation temperature, it is called *saturated vapor*.
- ❖ When the vapor is at a temperature greater than the saturation temperature, it is said to exist as *superheated vapor*.
- ❖ The pressure and temperature of superheated vapor are independent properties, since the temperature may increase while the pressure remains constant



Latent Heat

- ❖ **Latent heat:** The amount of energy absorbed or released during a phase-change process.
- ❖ **Latent heat of fusion:** The amount of energy absorbed during melting. It is equivalent to the amount of energy released during freezing.
- ❖ **Latent heat of vaporization:** The amount of energy absorbed during vaporization and it is equivalent to the energy released during condensation.
 - ❑ *At 1 atm pressure, the latent heat of fusion of water is 333.7 kJ/kg and the latent heat of vaporization is 2256.5 kJ/kg.*

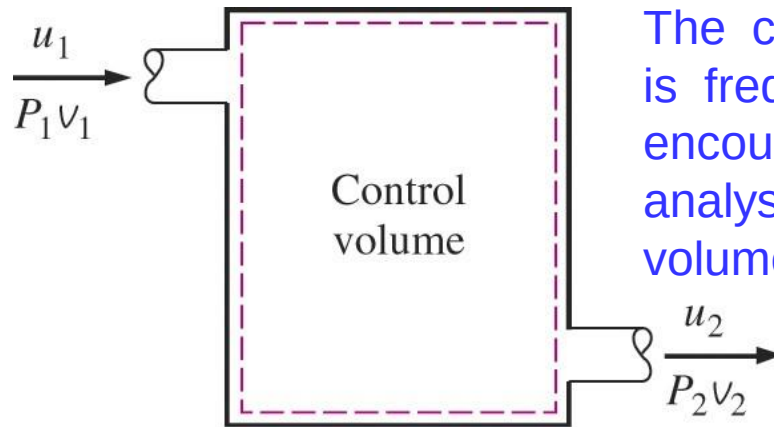
Property Table

- For most substances, the relationships among thermodynamic properties are too complex to be expressed by simple equations.
- Therefore, properties are frequently presented in the form of tables.
- Some thermodynamic properties can be measured easily, but others cannot and are calculated by using the relations between them and measurable properties.
- The results of these measurements and calculations are presented in tables in a convenient format.

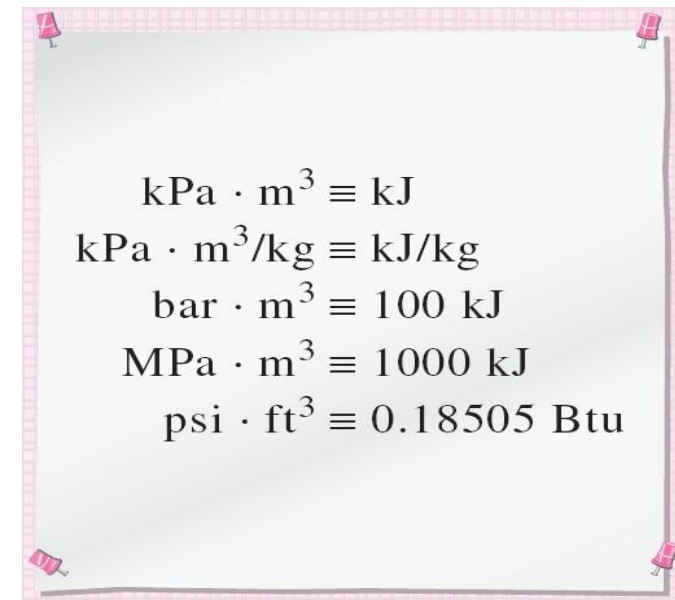
Enthalpy—A Combination Property

$$h = u + Pv \quad (\text{kJ/kg})$$

$$H = U + PV \quad (\text{kJ})$$



The combination $u + Pv$ is frequently encountered in the analysis of control volumes.



$\text{kPa} \cdot \text{m}^3 \equiv \text{kJ}$
$\text{kPa} \cdot \text{m}^3/\text{kg} \equiv \text{kJ/kg}$
$\text{bar} \cdot \text{m}^3 \equiv 100 \text{ kJ}$
$\text{MPa} \cdot \text{m}^3 \equiv 1000 \text{ kJ}$
$\text{psi} \cdot \text{ft}^3 \equiv 0.18505 \text{ Btu}$

The product *pressure* $\times$ *volume* has energy units.

Property Table

- ❖ For example if the pressure and specific volume are specified, three questions are asked: For the given pressure,

Temp. °C T	Sat. press. kPa P_{sat}	Specific volume m^3/kg	
		Sat. liquid ν_f	Sat. vapor ν_g
85	57.868	0.001032	2.8261
90	70.183	0.001036	2.3593
95	84.609	0.001040	1.9808

ν_f = specific volume of saturated liquid

ν_g = specific volume of saturated vapor

ν_{fg} = difference between ν_g and ν_f (that is, $\nu_{fg} = \nu_g - \nu_f$)

Specific temperature

Corresponding saturation pressure

Specific volume of saturated liquid

Specific volume of saturated vapor

Property Table

- ❖ If the answer to the first question is yes, the state is in the compressed liquid region, and the compressed liquid table is used to find the properties. (or using *saturation temperature table*)
- ❖ If the answer to the second question is yes, the state is in the saturation region, and either the saturation temperature table or the saturation pressure table is used.
- ❖ If the answer to the third question is yes, the state is in the **superheated region** and the superheated table is used.

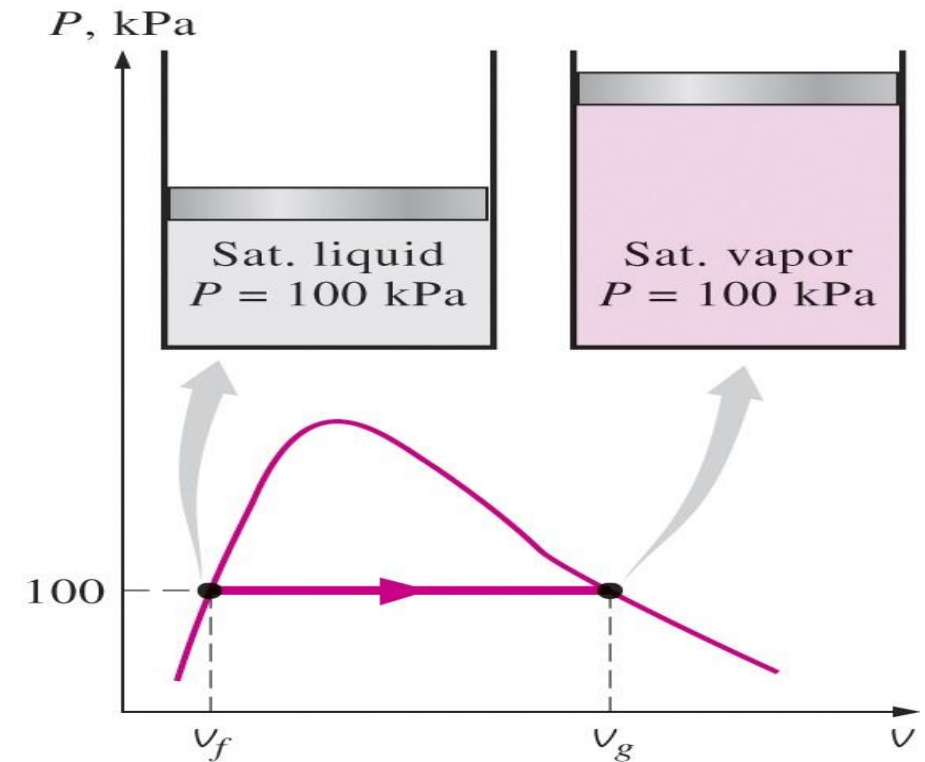
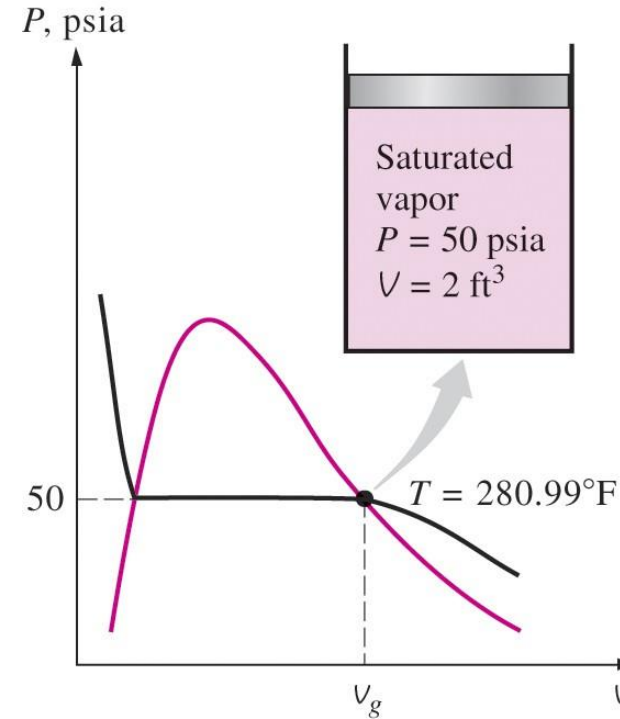
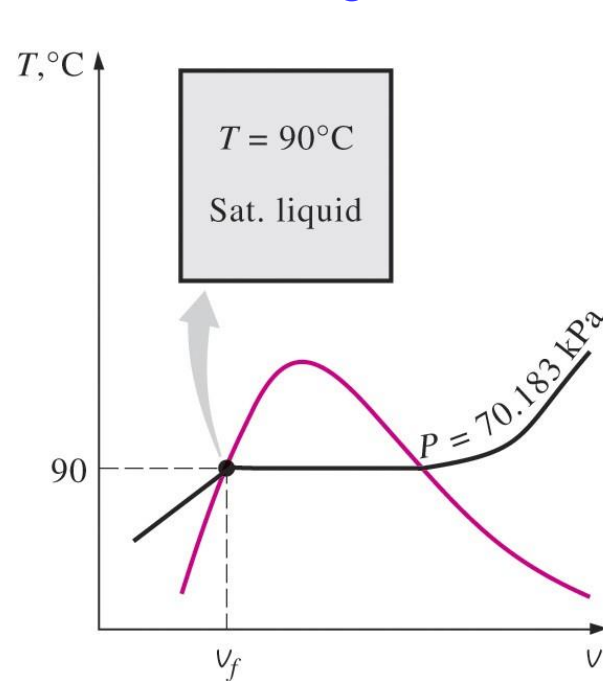
$$v < v_f \quad ?$$

$$v_f < v < v_g \quad ?$$

$$v_g < v \quad ?$$

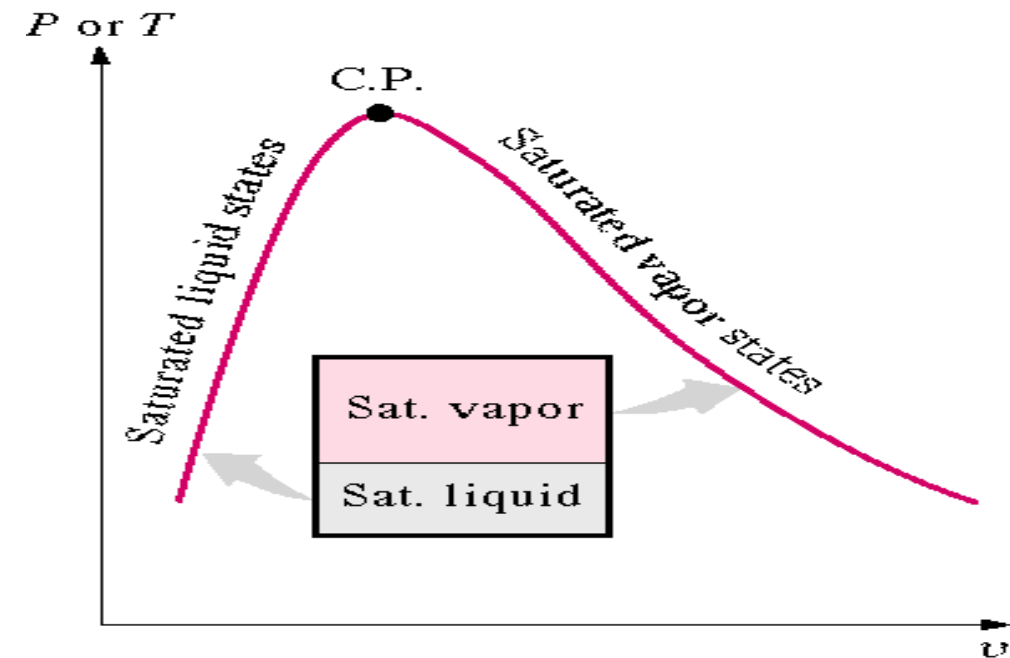
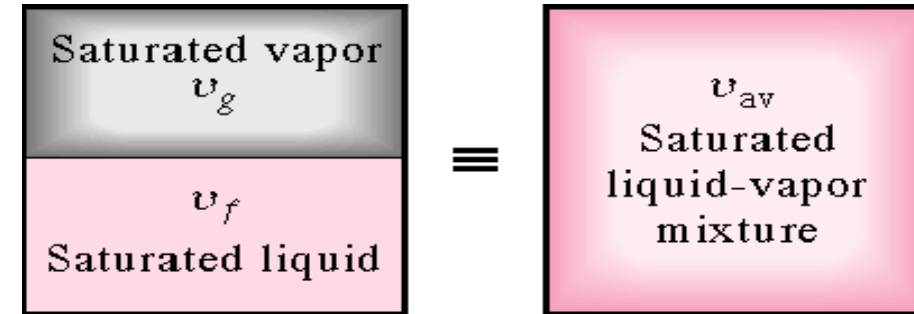
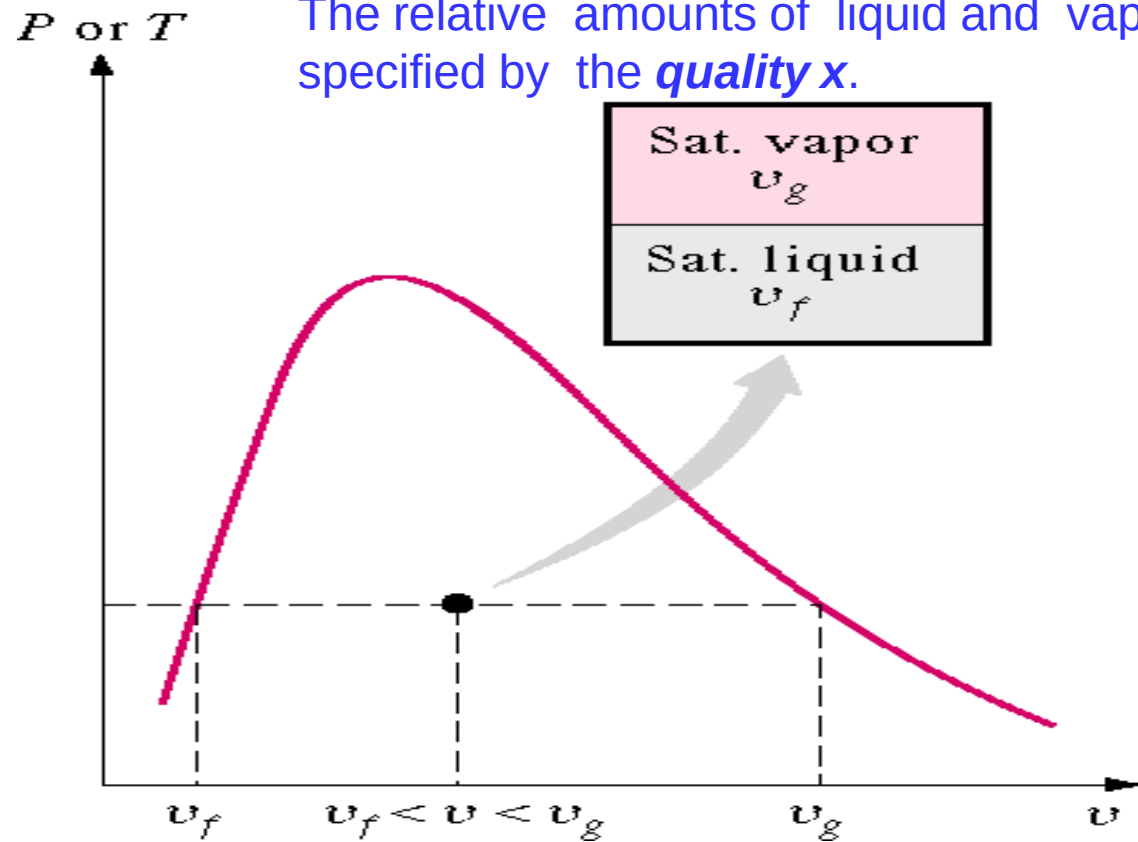
Property Table

Examples: Saturated liquid and saturated vapor states of water on T - v and P - v diagrams.



Saturated Liquid–Vapor Mixture

The relative amounts of liquid and vapor phases in a saturated mixture are specified by the *quality* x .





Quality

- ❖ When a substance exists as part liquid and part vapor at saturation conditions, its *quality* (x) is **defined** as the ratio of the mass of the vapor to the total mass of both vapor and liquid.
- ❖ The quality is zero for the saturated liquid and one for the saturated vapor ($0 \leq x \leq 1$)
- ❖ For example, if the mass of vapor is 0.2 g and the mass of the liquid is 0.8 g, then the quality is 0.2 or 20%.

$$x = \frac{\text{mass}_{\text{saturated vapor}}}{\text{mass}_{\text{total}}} = \frac{m_g}{m_f + m_g}$$

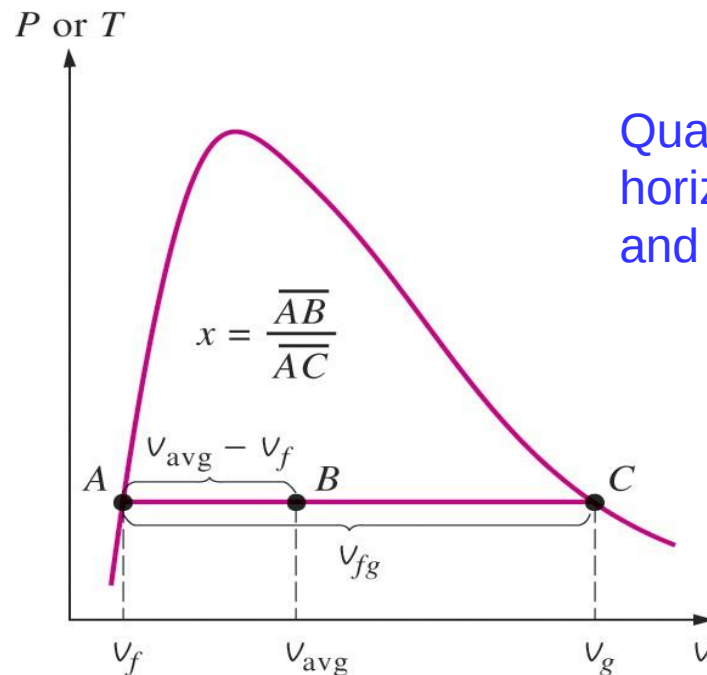
Quality

$$v_{\text{avg}} = v_f + x v_{fg} \quad (\text{m}^3/\text{kg})$$

$$x = \frac{v_{\text{avg}} - v_f}{v_{fg}}$$

$$u_{\text{avg}} = u_f + x u_{fg} \quad (\text{kJ}/\text{kg})$$

$$h_{\text{avg}} = h_f + x h_{fg} \quad (\text{kJ}/\text{kg})$$



Consider a tank that contains a saturated liquid–vapor mixture. The volume occupied by saturated liquid is V_f , and the volume occupied by saturated vapor is V_g . The total volume V is the sum of the two:

$$y \rightarrow v, u, \text{ or } h.$$

$$y_{\text{avg}} = y_f + x y_{fg}$$

$$y_f \leq y_{\text{avg}} \leq y_g$$

$$V = V_f + V_g$$

$$V = m v \rightarrow m_t v_{\text{avg}} = m_f v_f + m_g v_g$$

$$m_f = m_t - m_g \rightarrow m_t v_{\text{avg}} = (m_t - m_g) v_f + m_g v_g$$

Dividing by m_t yields

$$v_{\text{avg}} = (1 - x) v_f + x v_g$$

since $x = m_g / m_t$. This relation can also be expressed as

$$v_{\text{avg}} = v_f + x v_{fg} \quad (\text{m}^3/\text{kg})$$

where $v_{fg} = v_g - v_f$. Solving for quality, we obtain

$$x = \frac{v_{\text{avg}} - v_f}{v_{fg}}$$

The v value of a saturated liquid–vapor mixture lies between the v_f and v_g values at the specified T or P

Superheated Vapor

In the region to the right of the saturated vapor line and at temperatures above the critical point temperature, a substance exists as superheated vapor.

In this region, temperature and pressure are independent properties.

At a specified P , superheated vapor exists at a higher h than the saturated vapor

	v	u	h
$T, ^\circ\text{C}$	m^3/kg	kJ/kg	kJ/kg
$P = 0.1 \text{ MPa } (99.61^\circ\text{C})$			
Sat.	1.6941	2505.6	2675.0
100	1.6959	2506.2	2675.8
150	1.9367	2582.9	2776.6
$\vdots$	$\vdots$	$\vdots$	$\vdots$
1300	7.2605	4687.2	5413.3
$P = 0.5 \text{ MPa } (151.83^\circ\text{C})$			
Sat.	0.37483	2560.7	2748.1
200	0.42503	2643.3	2855.8
250	0.47443	2723.8	2961.0

Compared to saturated vapor, superheated vapor is characterized by

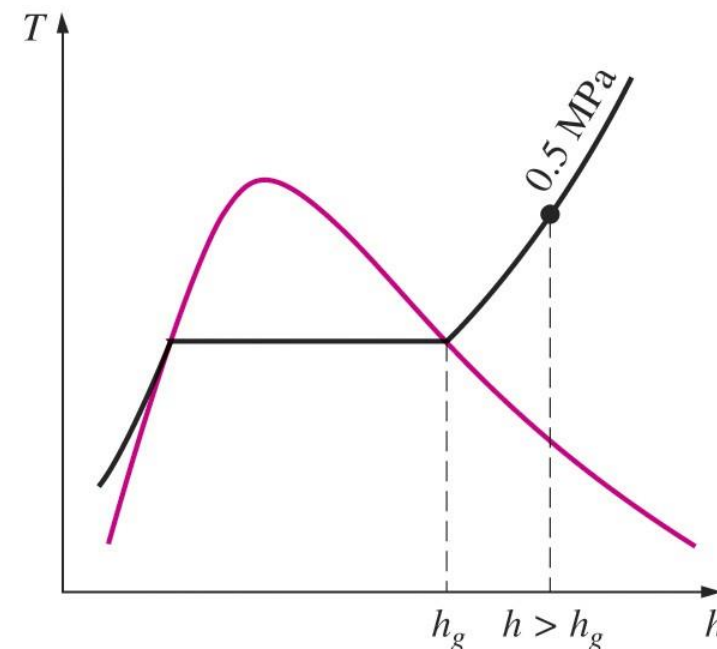
Lower pressures ($P < P_{\text{sat}}$ at a given T)

Higher temperatures ($T > T_{\text{sat}}$ at a given P)

Higher specific volumes ($v > v_g$ at a given P or T)

Higher internal energies ($u > u_g$ at a given P or T)

Higher enthalpies ($h > h_g$ at a given P or T)



Compressed Liquid

The compressed liquid properties depend on temperature much more strongly than they do on pressure.

$$y \cong y_f @ T \quad y \rightarrow v, u, \text{ or } h$$

Given: P and T

$$v \cong v_f @ T$$

$$u \cong u_f @ T$$

$$h \cong h_f @ T$$

At a given P and T , a pure substance will exist as a compressed liquid

A compressed liquid may be approximated as a saturated liquid at the given temperature if compressed liquid data is not given.

Compressed liquid is characterized by

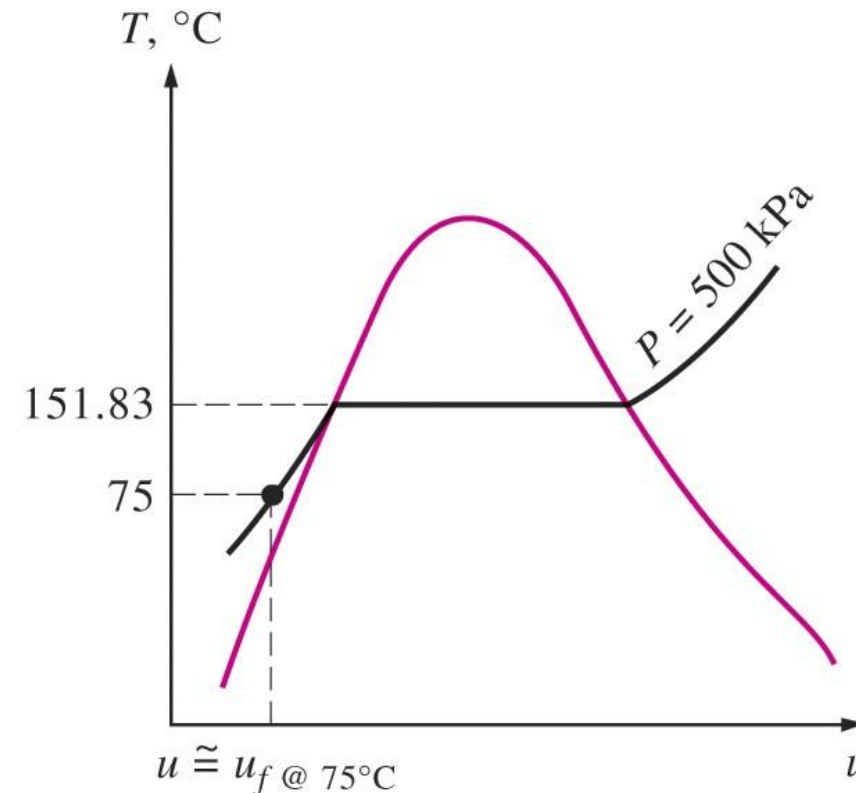
Higher pressures ($P > P_{\text{sat}}$ at a given T)

Lower temperatures ($T < T_{\text{sat}}$ at a given P)

Lower specific volumes ($v < v_f$ at a given P or T)

Lower internal energies ($u < u_f$ at a given P or T)

Lower enthalpies ($h < h_f$ at a given P or T)



Example 2.1

Determine the saturated pressure, specific volume, internal energy and enthalpy for saturated water vapor at 45°C and 50°C.

Saturated water—Temperature table

Temp., T °C	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg · K		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42.022	2477.2	2519.2	0.1511	8.7488	8.8999
15	1.7057	0.001001	77.885	62.980	2332.5	2395.5	62.982	2465.4	2528.3	0.2245	8.5559	8.7803
20	2.3392	0.001002	57.762	83.913	2318.4	2402.3	83.915	2453.5	2537.4	0.2965	8.3696	8.6661
25	3.1698	0.001003	43.340	104.83	2304.3	2409.1	104.83	2441.7	2546.5	0.3672	8.1895	8.5567
30	4.2469	0.001004	32.879	125.73	2290.2	2415.9	125.74	2429.8	2555.6	0.4368	8.0152	8.4520
35	5.6291	0.001006	25.205	146.63	2276.0	2422.7	146.64	2417.9	2564.6	0.5051	7.8466	8.3517
40	7.3851	0.001008	19.515	167.53	2261.9	2429.4	167.53	2406.0	2573.5	0.5724	7.6832	8.2556
45	9.5953	0.001010	15.251	188.43	2247.7	2436.1	188.44	2394.0	2582.4	0.6386	7.5247	8.1633
50	12.352	0.001012	12.026	209.33	2233.4	2442.7	209.34	2382.0	2591.3	0.7038	7.3710	8.0748
55	15.763	0.001015	9.5639	230.24	2219.1	2449.3	230.26	2369.8	2600.1	0.7680	7.2218	7.9898

Example 2.2

Determine the saturated pressure, specific volume, internal energy and enthalpy for saturated water vapor at 47° C .

Saturated water—Temperature table

Temp., T °C	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, $\text{kJ/kg} \cdot \text{K}$		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42.022	2477.2	2519.2	0.1511	8.7488	8.8999
15	1.7057	0.001001	77.885	62.980	2332.5	2395.5	62.982	2465.4	2528.3	0.2245	8.5559	8.7803
20	2.3392	0.001002	57.762	83.913	2318.4	2402.3	83.915	2453.5	2537.4	0.2965	8.3696	8.6661
25	3.1698	0.001003	43.340	104.83	2304.3	2409.1	104.83	2441.7	2546.5	0.3672	8.1895	8.5567
30	4.2469	0.001004	32.879	125.73	2290.2	2415.9	125.74	2429.8	2555.6	0.4368	8.0152	8.4520
35	5.6291	0.001006	25.205	146.63	2276.0	2422.7	146.64	2417.9	2564.6	0.5051	7.8466	8.3517
40	7.3851	0.001008	19.515	167.53	2261.9	2429.4	167.53	2406.0	2573.5	0.5724	7.6832	8.2556
45	9.5953	0.001010	15.251	188.43	2247.7	2436.1	188.44	2394.0	2582.4	0.6386	7.5247	8.1633
50	12.352	0.001012	12.026	209.33	2233.4	2442.7	209.34	2382.0	2591.3	0.7038	7.3710	8.0748
55	15.763	0.001015	9.5639	230.24	2219.1	2449.3	230.26	2369.8	2600.1	0.7680	7.2218	7.9898

Solution:

- Extract data from steam table

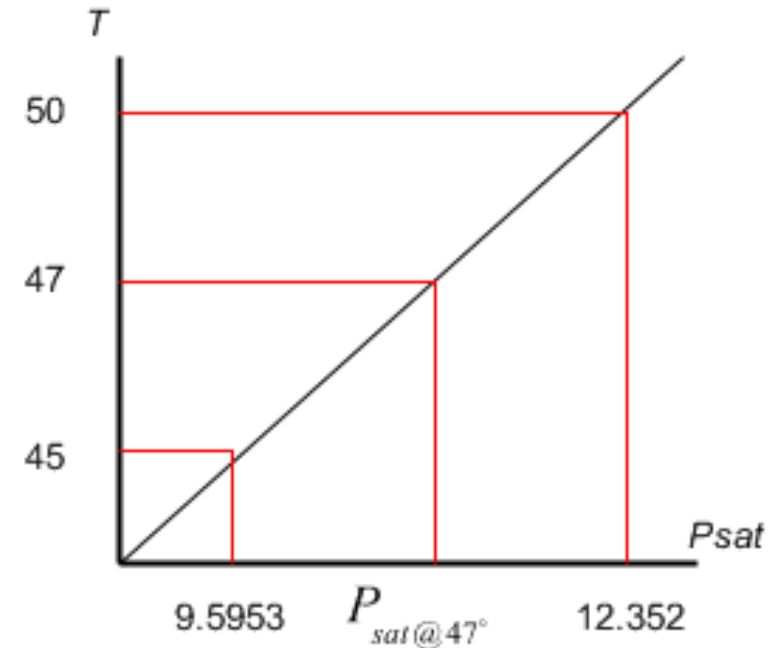
T	P_{sat}	v	u	h
45	9.5953	15.251	2436.1	2582.4
47	P_{sat}	v	u	h
50	12.352	12.026	2442.7	2591.3

- Interpolation for P_{sat}

$$\frac{P_{sat} - 9.5953}{12.352 - 9.5953} = \frac{47 - 45}{50 - 45}$$

$$P_{sat@47^\circ} = \underline{\underline{10.698 \text{ kPa}}}$$

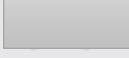
- Do the same principal to others!!!!

Interpolation Scheme for P_{sat}

Example 2.3

A rigid tank contains 50 kg of saturated liquid water at 90°C. Determine the pressure in the tank and the volume of the tank.

Solution A rigid tank contains saturated liquid water. The pressure and volume of the tank are to be determined.

Analysis The state of the saturated liquid water is shown on a T - v diagram in Fig. . Since saturation conditions exist in the tank, the pressure must be the saturation pressure at 90°C:

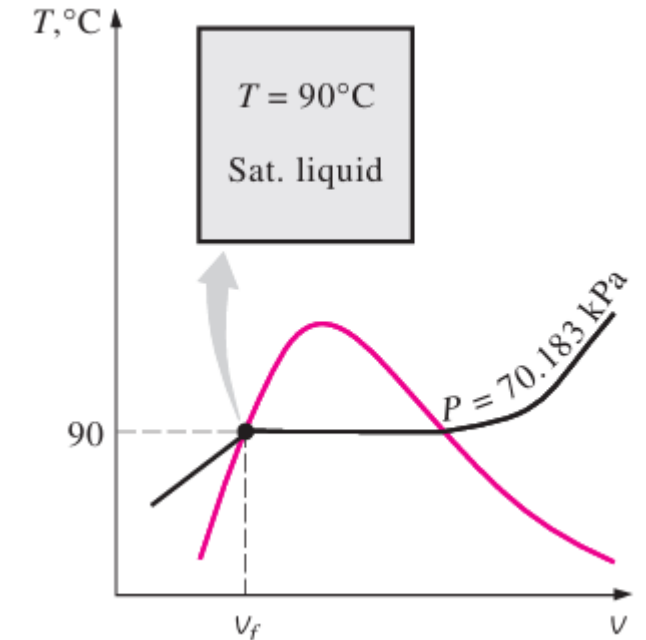
$$P = P_{\text{sat @ } 90^\circ\text{C}} = \mathbf{70.183 \text{ kPa}} \quad (\text{Table A-4})$$

The specific volume of the saturated liquid at 90°C is

$$v = v_f @ 90^\circ\text{C} = 0.001036 \text{ m}^3/\text{kg} \quad (\text{Table A-4})$$

Then the total volume of the tank becomes

$$V = m v = (50 \text{ kg})(0.001036 \text{ m}^3/\text{kg}) = \mathbf{0.0518 \text{ m}^3}$$



Schematic and T - v diagram for Example 2.3.

Example 2.4

A mass of 200 g of saturated liquid water is completely vaporized at a constant pressure of 100 kPa. Determine (a) the volume change and (b) the amount of energy transferred to the water.

Solution Saturated liquid water is vaporized at constant pressure. The volume change and the energy transferred are to be determined.

Analysis (a) The process described is illustrated on a P - v diagram in Fig.

The volume change per unit mass during a vaporization process is v_{fg} , which is the difference between v_g and v_f . Reading these values from Table A-5 at 100 kPa and substituting yield

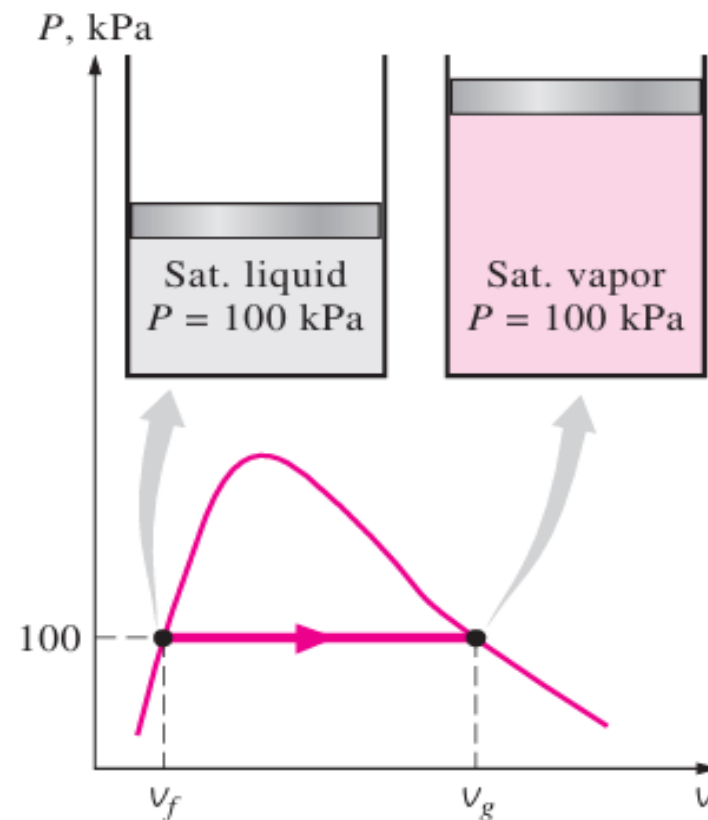
$$v_{fg} = v_g - v_f = 1.6941 - 0.001043 = 1.6931 \text{ m}^3/\text{kg}$$

Thus,

$$\Delta V = m v_{fg} = (0.2 \text{ kg})(1.6931 \text{ m}^3/\text{kg}) = \mathbf{0.3386 \text{ m}^3}$$

(b) The amount of energy needed to vaporize a unit mass of a substance at a given pressure is the enthalpy of vaporization at that pressure, which is $h_{fg} = 2257.5 \text{ kJ/kg}$ for water at 100 kPa. Thus, the amount of energy transferred is

$$m h_{fg} = (0.2 \text{ kg})(2257.5 \text{ kJ/kg}) = \mathbf{451.5 \text{ kJ}}$$



Schematic and P - v diagram for Example 2.4.

Example 2.5

A rigid tank contains 10 kg of water at 90°C. If 8 kg of the water is in the liquid form and the rest is in the vapor form, determine (a) the pressure in the tank and (b) the volume of the tank.

Solution A rigid tank contains saturated mixture. The pressure and the volume of the tank are to be determined.

Analysis (a) The state of the saturated liquid–vapor mixture is shown in Fig. Since the two phases coexist in equilibrium, we have a saturated mixture, and the pressure must be the saturation pressure at the given temperature:

$$P = P_{\text{sat @ } 90^\circ\text{C}} = \mathbf{70.183 \text{ kPa}} \quad (\text{Table A-4})$$

(b) At 90°C, we have $\nu_f = 0.001036 \text{ m}^3/\text{kg}$ and $\nu_g = 2.3593 \text{ m}^3/\text{kg}$ (Table A-4). One way of finding the volume of the tank is to determine the volume occupied by each phase and then add them:

$$\begin{aligned} V &= V_f + V_g = m_f \nu_f + m_g \nu_g \\ &= (8 \text{ kg})(0.001036 \text{ m}^3/\text{kg}) + (2 \text{ kg})(2.3593 \text{ m}^3/\text{kg}) \\ &= \mathbf{4.73 \text{ m}^3} \end{aligned}$$

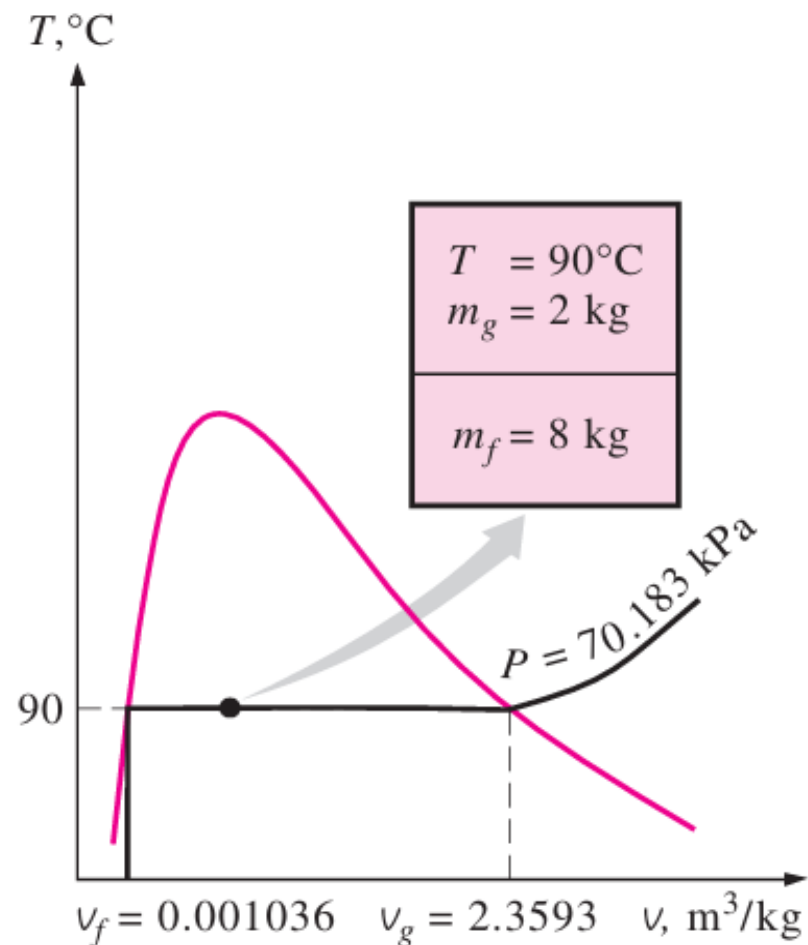
Another way is to first determine the quality x , then the average specific volume ν , and finally the total volume:

$$x = \frac{m_g}{m_t} = \frac{2 \text{ kg}}{10 \text{ kg}} = 0.2$$

$$\begin{aligned} \nu &= \nu_f + x \nu_{fg} \\ &= 0.001036 \text{ m}^3/\text{kg} + (0.2)[(2.3593 - 0.001036) \text{ m}^3/\text{kg}] \\ &= 0.473 \text{ m}^3/\text{kg} \end{aligned}$$

and

$$V = m\nu = (10 \text{ kg})(0.473 \text{ m}^3/\text{kg}) = 4.73 \text{ m}^3$$

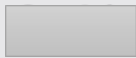


Schematic and T-v diagram for Example 2.5.

Example 2.6

Determine the temperature of water at a state of $P = 0.5 \text{ MPa}$ and $h = 2890 \text{ kJ/kg}$.

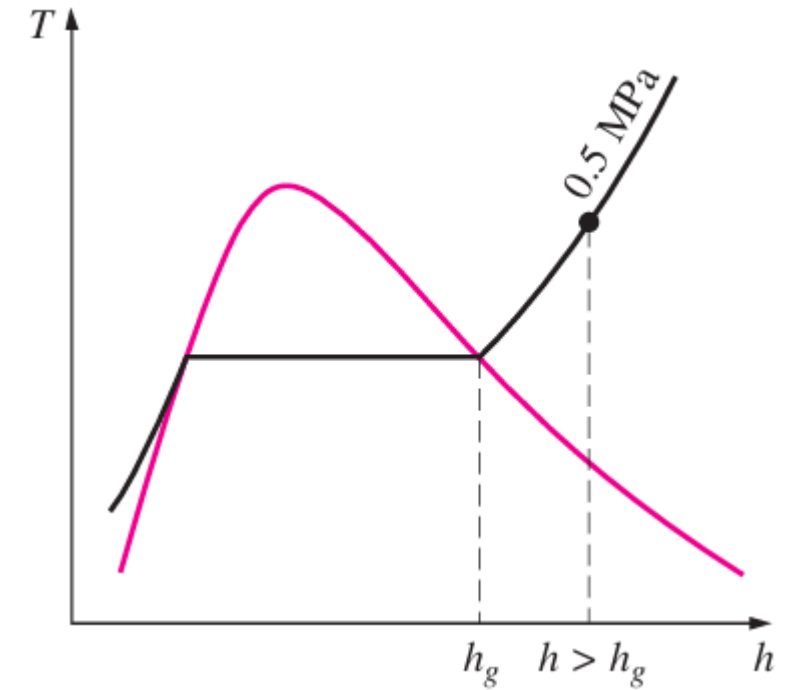
Solution The temperature of water at a specified state is to be determined.

Analysis At 0.5 MPa , the enthalpy of saturated water vapor is $h_g = 2748.1 \text{ kJ/kg}$. Since $h > h_g$, as shown in Fig.  we again have superheated vapor. Under 0.5 MPa in Table A-6 we read

$T, ^\circ\text{C}$	$h, \text{kJ/kg}$
200	2855.8
250	2961.0

Obviously, the temperature is between 200 and 250°C . By linear interpolation it is determined to be

$$T = 216.3^\circ\text{C}$$



At a specified P , superheated vapor exists at a higher h than the saturated vapor (Example 2.6).

Example 2.7

Determine the internal energy of compressed liquid water at 80°C and 5 MPa, using (a) data from the compressed liquid table and (b) saturated liquid data. What is the error involved in the second case?

Solution The exact and approximate values of the internal energy of liquid water are to be determined.

Analysis At 80°C, the saturation pressure of water is 47.416 kPa, and since $5 \text{ MPa} > P_{\text{sat}}$, we obviously have compressed liquid, as shown in Fig. 

(a) From the compressed liquid table (Table A-7)

$$\left. \begin{array}{l} P = 5 \text{ MPa} \\ T = 80^\circ\text{C} \end{array} \right\} u = \mathbf{333.82 \text{ kJ/kg}}$$

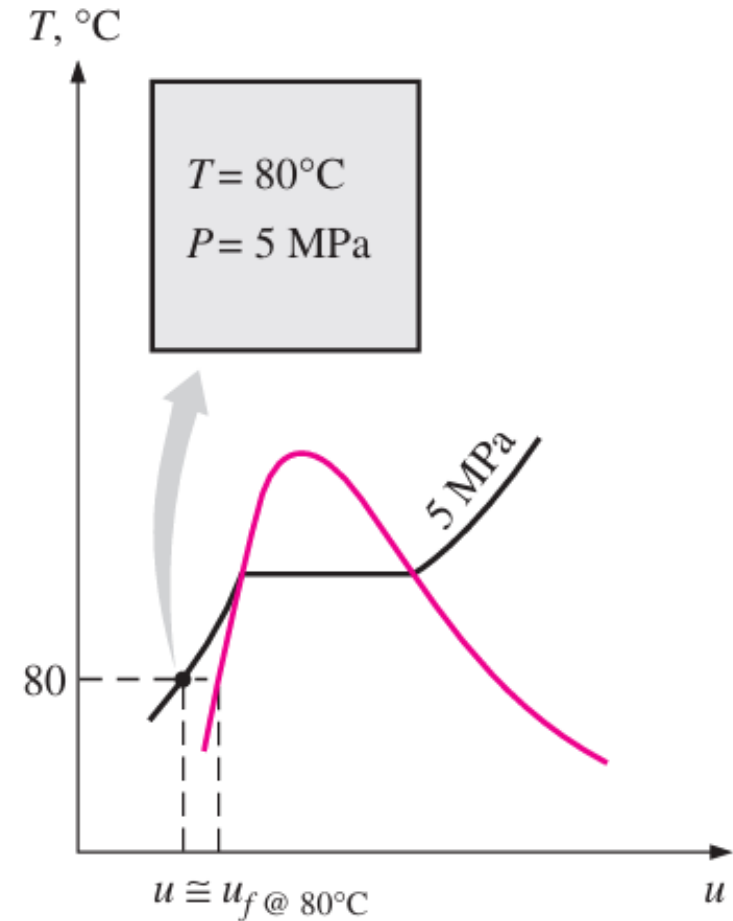
(b) From the saturation table (Table A-4), we read

$$u \cong u_f @ 80^\circ\text{C} = \mathbf{334.97 \text{ kJ/kg}}$$

The error involved is

$$\frac{334.97 - 333.82}{333.82} \times 100 = \mathbf{0.34\%}$$

which is less than 1 percent.



Schematic and T-u diagram for Example 2.7.



Exercises

1. Show the necessary procedure and Complete this table for H₂O:

$T, ^\circ\text{C}$	P, kPa	$h, \text{kJ/kg}$	x	Phase description
	200		0.7	
140		1800		
	950		0.0	
80	500			
	800	3162.2		

- Determine the saturated temperature, saturated pressure and enthalpy for water at specific volume of saturated vapor at $10.02 \text{ m}^3/\text{kg}$.
- Steam is contained in a closed rigid container with a volume of 1 m^3 . Initially, the pressure and temperature of the steam are 7 bar and 500°C , respectively. The temperatures drop as a result of heat transfer to the surroundings. Determine the temperature at which condensation first occurs, in $^\circ\text{C}$, and the fraction of the total mass that has condensed when the pressure reaches 0.5 bar. What is the volume, in m^3 , occupied by saturated liquid at the final state?



Example 2.8

Determine the enthalpy of 1.5 kg of water contained in a volume of 1.2 m³ at 200 kPa.

Solution:

□ Specific volume for water

$$v = \frac{\text{Volume}}{\text{mass}} = \frac{1.2 \text{ m}^3}{1.5 \text{ kg}} = 0.8 \frac{\text{m}^3}{\text{kg}}$$

□ From table A-5:

$$v_f = 0.001061 \frac{\text{m}^3}{\text{kg}}$$

$$v_g = 0.8858 \frac{\text{m}^3}{\text{kg}}$$

Is $v < v_f$? No

Is $v_f < v < v_g$? Yes

Is $v_g < v$? No

□ Find the quality

$$v = v_f + x(v_g - v_f)$$

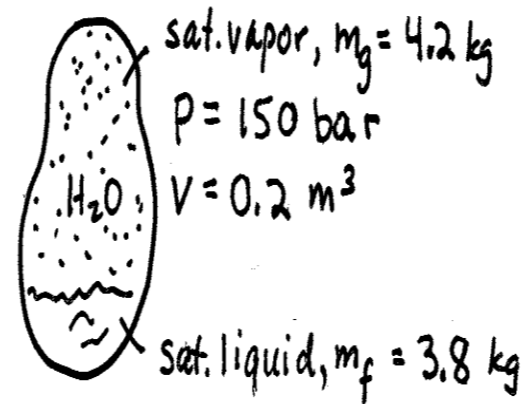
$$\begin{aligned} x &= \frac{v - v_f}{v_g - v_f} \\ &= \frac{0.8 - 0.001061}{0.8858 - 0.001061} \\ &= 0.903 \quad (\text{What does this mean?}) \end{aligned}$$

□ The enthalpy

$$\begin{aligned} h &= h_f + x h_{fg} \\ &= 504.7 + (0.903)(2201.6) \\ &= 2492.7 \frac{\text{kJ}}{\text{kg}} \end{aligned}$$

Example 2.9

A two-phase liquid–vapor mixture of a substance has a pressure of 150 bar and occupies a volume of 0.2 m^3 of saturated liquid and vapor present are 3.8 kg and 4.2 kg, respectively Determine the mixture specific volume in m^3/kg .



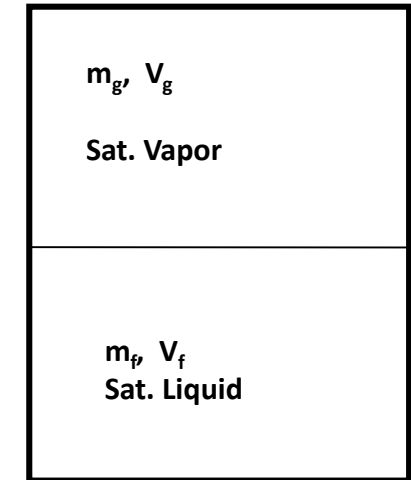
For the two-phase liquid-vapor mixture, the specific volume is

$$\begin{aligned}
 v &= \frac{V}{m_f + m_g} \\
 &= \frac{0.2}{(3.8 + 4.2)} \\
 &= 0.025 \text{ m}^3/\text{kg} \leftarrow v
 \end{aligned}$$



Example 2.10

Consider the closed, rigid container of water as shown. The pressure is 700 kPa, the mass of the saturated liquid is 1.78 kg, and the mass of the saturated vapor is 0.22 kg. Heat is added to the water until the pressure increases to 8 MPa. Find the final temperature, enthalpy, and internal energy of the water



Solution:

- ❖ Theoretically:

$$v_2 = v_1$$

- ❖ The quality before pressure increased (**state 1**).

$$\begin{aligned} x_1 &= \frac{m_{g1}}{m_{f1} + m_{g1}} \\ &= \frac{0.22 \text{ kg}}{(1.78 + 0.22) \text{ kg}} = 0.11 \end{aligned}$$

- ❖ Specific volume at **state 1**

$$\begin{aligned} v_1 &= v_{f1} + x_1 (v_{g1} - v_{f1}) \\ &= 0.001108 + (0.11)(0.2728 - 0.001108) \\ &= 0.031 \frac{\text{m}^3}{\text{kg}} \end{aligned}$$

State 2:

- ❖ Information :

$$P_2 = 8 \text{ MPa} \quad v_2 = 0.031 \frac{\text{m}^3}{\text{kg}}$$

- ❖ From table A-5:

$$\begin{aligned} v_{f,2} &= 0.001384 \frac{\text{m}^3}{\text{kg}} \\ v_{g,2} &= 0.02352 \frac{\text{m}^3}{\text{kg}} \end{aligned}$$

$v_{g2} < v_2$

- ❖ Since that it is in superheated region, use table A-6:

$$T_2 = 361.8^\circ \text{C}$$

$$h_2 = 3024 \frac{\text{kJ}}{\text{kg}}$$

$$u_2 = 2776 \frac{\text{kJ}}{\text{kg}}$$



Reference State and Reference Values

- The values of u , h , and s cannot be measured directly, and they are calculated from measurable properties using the relations between properties.
- However, those relations give the *changes* in properties, not the values of properties at specified states.
- Therefore, we need to choose a convenient *reference state* and assign a value of *zero* for a convenient property or properties at that state.
- The reference state for water is 0.01°C and for R-134a is -40°C in tables.
- Some properties may have negative values as a result of the reference state chosen.
- Sometimes different tables list different values for some properties at the same state as a result of using a different reference state.
- However, In thermodynamics we are concerned with the *changes* in properties, and the reference state chosen is of no consequence in calculations.

Saturated water—Temperature table

Temp., T °C	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, $\text{kJ/kg} \cdot \text{K}$		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249

Saturated refrigerant-134a—Temperature table

Temp., T °C	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, $\text{kJ/kg} \cdot \text{K}$		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
-40	51.25	0.0007054	0.36081	-0.036	207.40	207.37	0.000	225.86	225.86	0.00000	0.96866	0.96866

THE IDEAL-GAS EQUATION OF STATE

- **Equation of state:** Any equation that relates the pressure, temperature, and specific volume of a substance.
- The simplest and best-known equation of state for substances in the gas phase is the ideal-gas equation of state. This equation predicts the P - v - T behavior of a gas quite accurately within some properly selected region.

- **Ideal gas equation of state**

$$Pv = RT \quad P = R \left(\frac{T}{v} \right)$$

$$R = \frac{R_u}{M} \quad (\text{kJ/kg} \cdot \text{K} \text{ or } \text{kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})$$

R : gas constant

M : molar mass (kg/kmol)

R_u : universal gas constant

$$R_u = \begin{cases} 8.31447 \text{ kJ/kmol} \cdot \text{K} \\ 8.31447 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K} \\ 0.0831447 \text{ bar} \cdot \text{m}^3/\text{kmol} \cdot \text{K} \\ 1.98588 \text{ Btu/lbmol} \cdot \text{R} \\ 10.7316 \text{ psia} \cdot \text{ft}^3/\text{lbmol} \cdot \text{R} \\ 1545.37 \text{ ft} \cdot \text{lbf/lbmol} \cdot \text{R} \end{cases}$$

Substance	R , kJ/kg·K
Air	0.2870
Helium	2.0769
Argon	0.2081
Nitrogen	0.2968

Different substances have different gas constants.

Mass = Molar mass $\times$ Mole number

$$m = MN \quad (\text{kg})$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Ideal gas equation at two states for a fixed mass

$$V = m v \longrightarrow P V = m R T$$

$$m R = (M N) R = N R_u \longrightarrow P V = N R_u T$$

Various expressions of ideal gas equation

Real gases behave as an ideal gas at low densities (i.e., low pressure, high temperature).

$$V = N \bar{v} \longrightarrow P \bar{v} = R_u T$$

Per unit mass	Per unit mole
$v, \text{m}^3/\text{kg}$	$\bar{v}, \text{m}^3/\text{kmol}$
$u, \text{kJ}/\text{kg}$	$\bar{u}, \text{kJ}/\text{kmol}$
$h, \text{kJ}/\text{kg}$	$\bar{h}, \text{kJ}/\text{kmol}$

The ideal-gas relation often is not applicable to real gases; thus, care should be exercised when using it.

Properties per unit mole are denoted with a bar on the top.



Ideal Gas Law

- ❖ *Robert Boyle* formulates a well-known law that states the pressure of a gas expanding at constant temperature varies inversely to the volume, or

$$P_1V_1 = P_2V_2 = \text{constant}$$

- ❖ As the result of experimentation, *Charles* concluded that the pressure of a gas varies directly with temperature when the volume is held constant, and the volume varies directly with temperature when the pressure is held constant, or

$$\frac{V_1}{V_2} = \frac{T_1}{T_2} \quad \text{or} \quad \frac{P_1}{P_2} = \frac{T_1}{T_2}$$



- ❖ By combining the results of Charles' and Boyle's experiments, the following relationship can be obtained
- ❖ The constant in the above equation is called *the ideal gas constant* and is designated by R ; thus the ideal gas equation becomes
- ❖ In order to make the equation applicable to all ideal gas, a universal gas constant R_U is introduced

$$\frac{Pv}{T} = \text{constant}$$

$$Pv = RT \quad \text{or} \quad PV = mRT$$

$$R = \frac{R_U}{M}$$



- ❑ For example the ideal gas constant for air, R_{air}

$$R_{\text{air}} = \frac{(R_U)_{\text{air}}}{(M)_{\text{air}}} = \frac{8.3144}{28.96} = 0.2871 \text{ kJ} / \text{kg.K}$$

- ❑ The amount of energy needed to raise the temperature of a unit of mass of a substance by one degree is called the *specific heat at constant volume* C_v for a constant-volume process and the *specific heat at constant pressure* C_p for a constant pressure process. They are defined as

$$C_v = \left(\frac{\partial u}{\partial T} \right)_v \quad \text{and} \quad C_p = \left(\frac{\partial h}{\partial T} \right)_p$$



- Using the definition of enthalpy ($h = u + Pv$) and writing the differential of enthalpy, the relationship between the specific heats for ideal gases is

$$h = u + Pv$$

$$dh = du + RT$$

$$C_P dt = C_V dt + R dT$$

$$C_P = C_V + R$$

- The *specific heat ratio*, k is defined as

$$k = \frac{C_P}{C_V}$$



- For ideal gases u , h , C_v , and C_p are functions of temperature alone.
The Δu and Δh of ideal gases can be expressed as

$$\Delta u = u_2 - u_1 = C_v(T_2 - T_1)$$

$$\Delta h = h_2 - h_1 = C_p(T_2 - T_1)$$

Example 2.11

An ideal gas is contained in a closed assembly with an initial pressure and temperature of 220 kPa and 70°C respectively. If the volume of the system is increased 1.5 times and the temperature drops to 15°C, determine the final pressure of the gas.

Solution:

given

state 1

$$P_1 = 220 \text{ kPa}$$

$$T_1 = 70 + 273 \text{ K} = 343 \text{ K}$$

state 2

$$T_2 = 15 + 273 = 288 \text{ K}$$

$$V_2 = 1.5V_1$$

❖ From ideal-gas law:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\begin{aligned} P_2 &= \frac{V_1}{1.5V_1} \left(\frac{288}{343} \right) (220 \times 10^3) \\ &= 123.15 \text{ kPa} \end{aligned}$$



Example 2.12

A closed assembly contains 2 kg of air at an initial pressure and temperature of 140 kPa and 210°C respectively. If the volume of the system is doubled and temperature drops to 37°C, determine the final pressure of the air. Air can be modeled as an ideal gas.

Solution:

given

state 1

$$P_1 = 140 \text{ kPa}$$

$$T_1 = 210 + 273 \text{ K} = 483 \text{ K}$$

state 2

$$T_2 = 37 + 273 = 310 \text{ K}$$

$$V_2 = 2V_1$$

❖ From ideal-gas law:

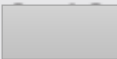
$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\begin{aligned} P_2 &= \frac{V_1}{2V_1} \left(\frac{310}{483} \right) (140 \times 10^3) \\ &= 44.93 \text{ kPa} \end{aligned}$$

Example 2.13

Determine the mass of the air in a room whose dimensions are 4 m $\times$ 5 m $\times$ 6 m at 100 kPa and 25°C.

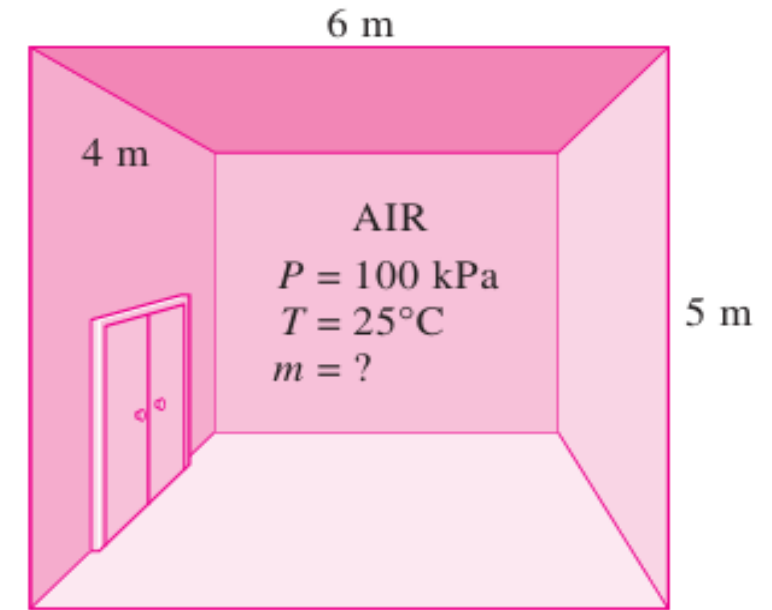
Solution The mass of air in a room is to be determined.

Analysis A sketch of the room is given in Fig. . Air at specified conditions can be treated as an ideal gas. From Table A-1, the gas constant of air is $R = 0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}$, and the absolute temperature is $T = 25^\circ\text{C} + 273 = 298 \text{ K}$. The volume of the room is

$$V = (4 \text{ m})(5 \text{ m})(6 \text{ m}) = 120 \text{ m}^3$$

The mass of air in the room is determined from the ideal-gas relation to be

$$m = \frac{PV}{RT} = \frac{(100 \text{ kPa})(120 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(298 \text{ K})} = \mathbf{140.3 \text{ kg}}$$



Schematic for Example 2.13.

Two Forms of the Virial Equation

A useful auxiliary thermodynamic property is defined by the equation:

$$Z \equiv \frac{PV}{RT}$$

This dimensionless ratio is called the **compressibility factor**. With this definition and with $a = RT \left[(PV)^* \right] = RT$]

,Eq. $PV = a(1 + B'P + C'P^2 + D'P^3 + \dots)$ becomes: $Z = 1 + B'P + C'P^2 + D'P^3 + \dots$

An alternative expression for Z is also in common use

$$Z = 1 + \frac{B}{v} + \frac{C}{v^2} + \frac{D}{v^3} + \dots$$

Both of these equations are known as virial expansions, and the parameters B' , C' , D' , etc., and B , C , D , etc., are called virial coefficients. Parameters B' and B are second virial coefficients; C' and C are third virial coefficients; etc. For a given gas the virial coefficients are functions of temperature only.

$$B' = \frac{B}{RT}$$

$$C' = \frac{C - B^2}{(RT)^2}$$

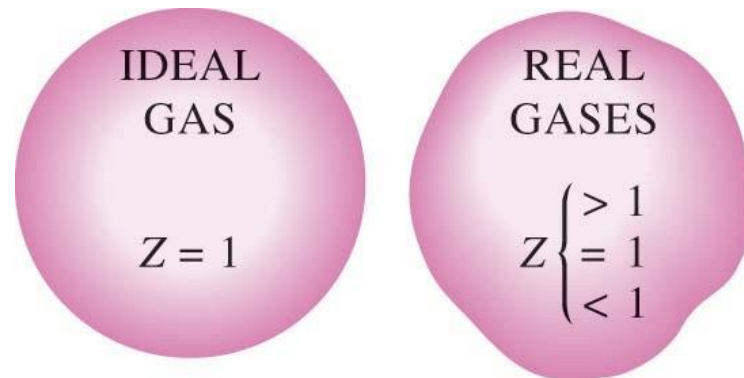
$$D' = \frac{D - 3BC + 2B^3}{(RT)^3}$$

COMPRESSIBILITY FACTOR—A MEASURE OF DEVIATION FROM IDEAL-GAS BEHAVIOR

Compressibility factor Z A factor that accounts for the deviation of real gases from ideal-gas behavior at a given temperature and pressure.

$$PV = ZRT$$

$$Z = \frac{PV}{RT} \quad Z = \frac{V_{\text{actual}}}{V_{\text{ideal}}}$$



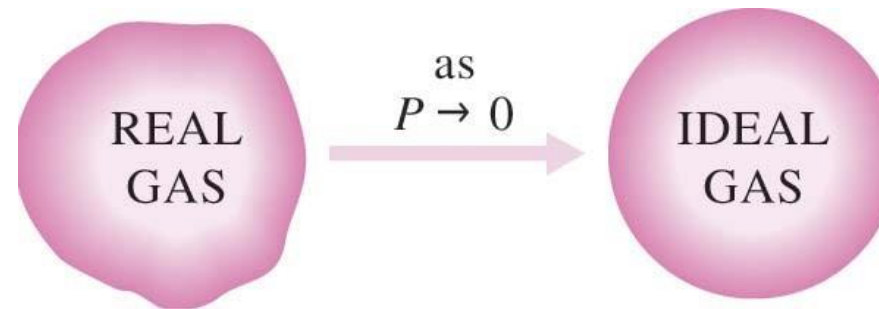
The compressibility factor is unity for ideal gases.

The farther away Z is from unity, the more the gas deviates from ideal-gas behavior.

Gases behave as an ideal gas at low densities (i.e., low pressure, high temperature).

Question: What is the criteria for low pressure and high temperature?

Answer: The pressure or temperature of a gas is high or low relative to its critical temperature or pressure.



At very low pressures, all gases approach ideal-gas behavior (regardless of their temperature).

$$P_R = \frac{P}{P_{cr}}$$

Reduced
pressure

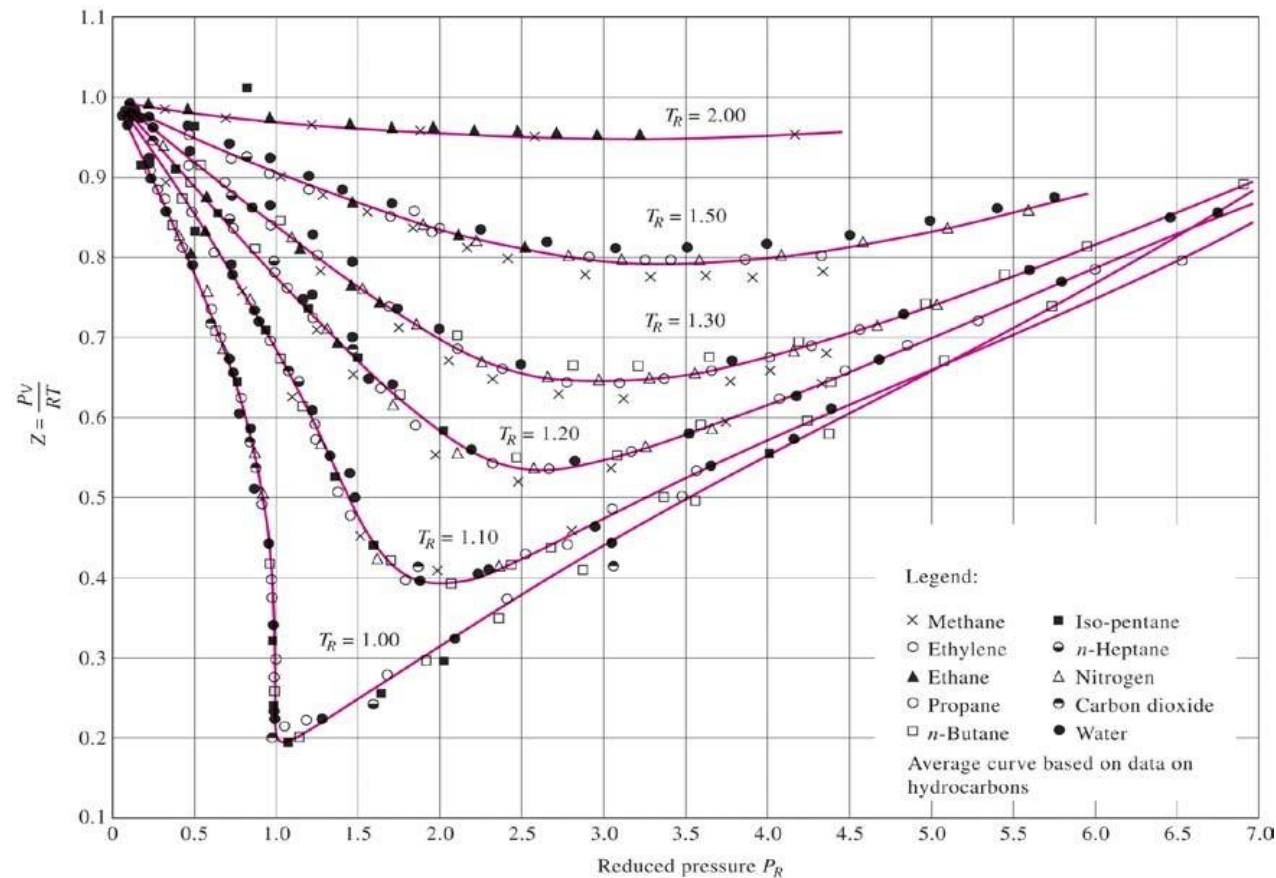
Reduced
temperature

$$T_R = \frac{T}{T_{cr}}$$

Pseudo-reduced specific
volume

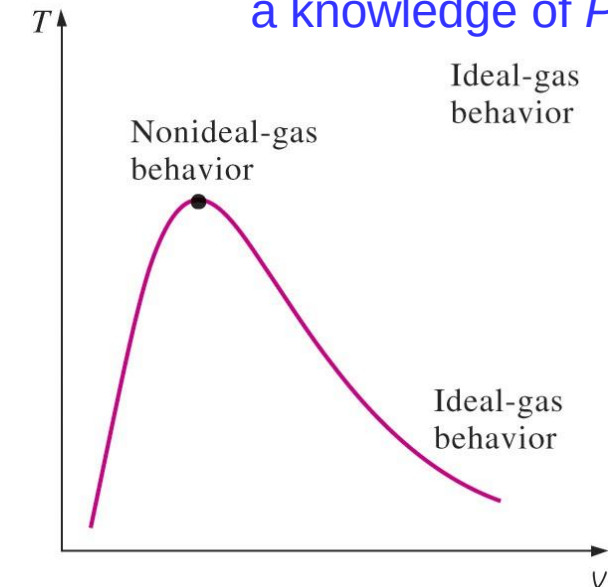
$$v_R = \frac{v_{actual}}{RT_{cr}/P_{cr}}$$

$$\left. \begin{aligned} P_R &= \frac{P}{P_{cr}} \\ v_R &= \frac{v}{RT_{cr}/P_{cr}} \end{aligned} \right\} Z = \dots \quad (\text{Fig. A-15})$$



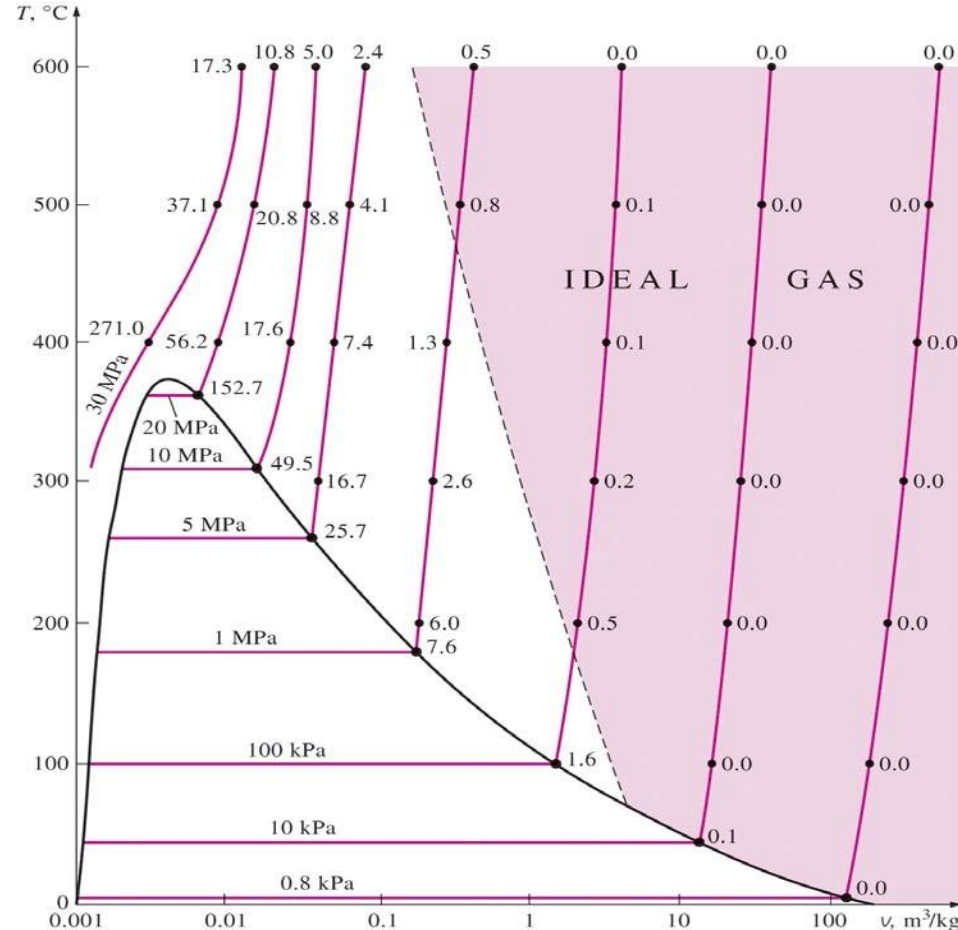
Comparison of Z factors for various gases.

Z can also be determined from
a knowledge of P_R and v_R .



Gases deviate from the ideal-gas
behavior the most in the neighborhood
of the critical point.

Is Water Vapor an Ideal Gas?



- At pressures below 10 kPa, water vapor can be treated as an ideal gas, regardless of its temperature, with negligible error (less than 0.1 percent).
- At higher pressures, however, the ideal gas assumption yields unacceptable errors, particularly in the vicinity of the critical point and the saturated vapor line.
- In air-conditioning applications, the water vapor in the air can be treated as an ideal gas. Why?
- In steam power plant applications, however, the pressures involved are usually very high; therefore, ideal-gas relations should not be used.

Percentage of error ($[(v_{\text{table}} - v_{\text{ideal}})/v_{\text{table}}] \times 100$) involved in assuming steam to be an ideal gas, and the region where steam can be treated as an ideal gas with less than 1 percent error.

CUBIC EQUATIONS OF STATE

Several equations have been proposed to represent the P - v - T behavior of substances accurately over a larger region with no limitations.

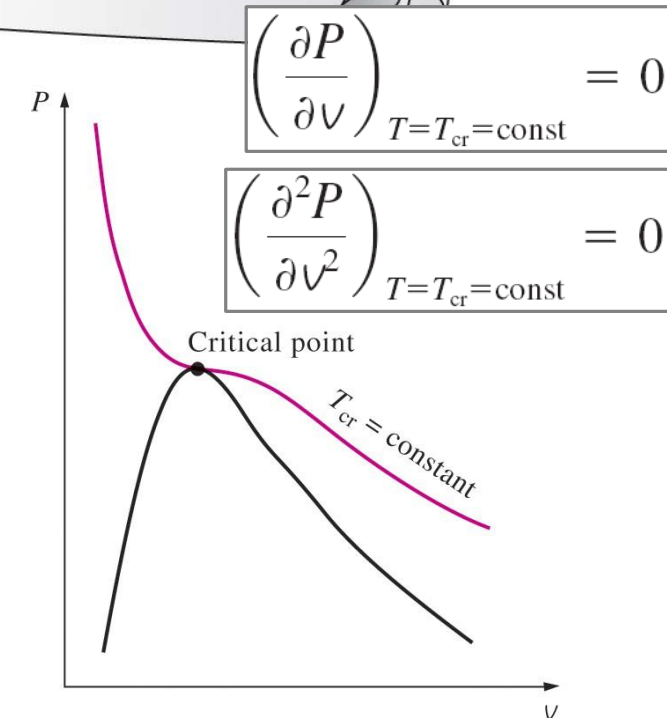
Van der Waals Equation of State

$$\left(P + \frac{a}{v^2}\right)(v - b) = RT$$

$$a = \frac{27R^2T_{cr}^2}{64P_{cr}} \quad b = \frac{RT_{cr}}{8P_{cr}}$$

Critical isotherm of a pure substance has an inflection point at the critical state.

This model includes two effects not considered in the ideal-gas model: *the intermolecular attraction forces and the volume occupied by the molecules themselves*. The accuracy of the van der Waals equation of state is often inadequate.





Beattie-Bridgeman Equation of State

$$P = \frac{R_u T}{\bar{V}^2} \left(1 - \frac{c}{\bar{V} T^3} \right) (\bar{V} + B) - \frac{A}{\bar{V}^2}$$

$$A = A_0 \left(1 - \frac{a}{\bar{V}} \right)$$

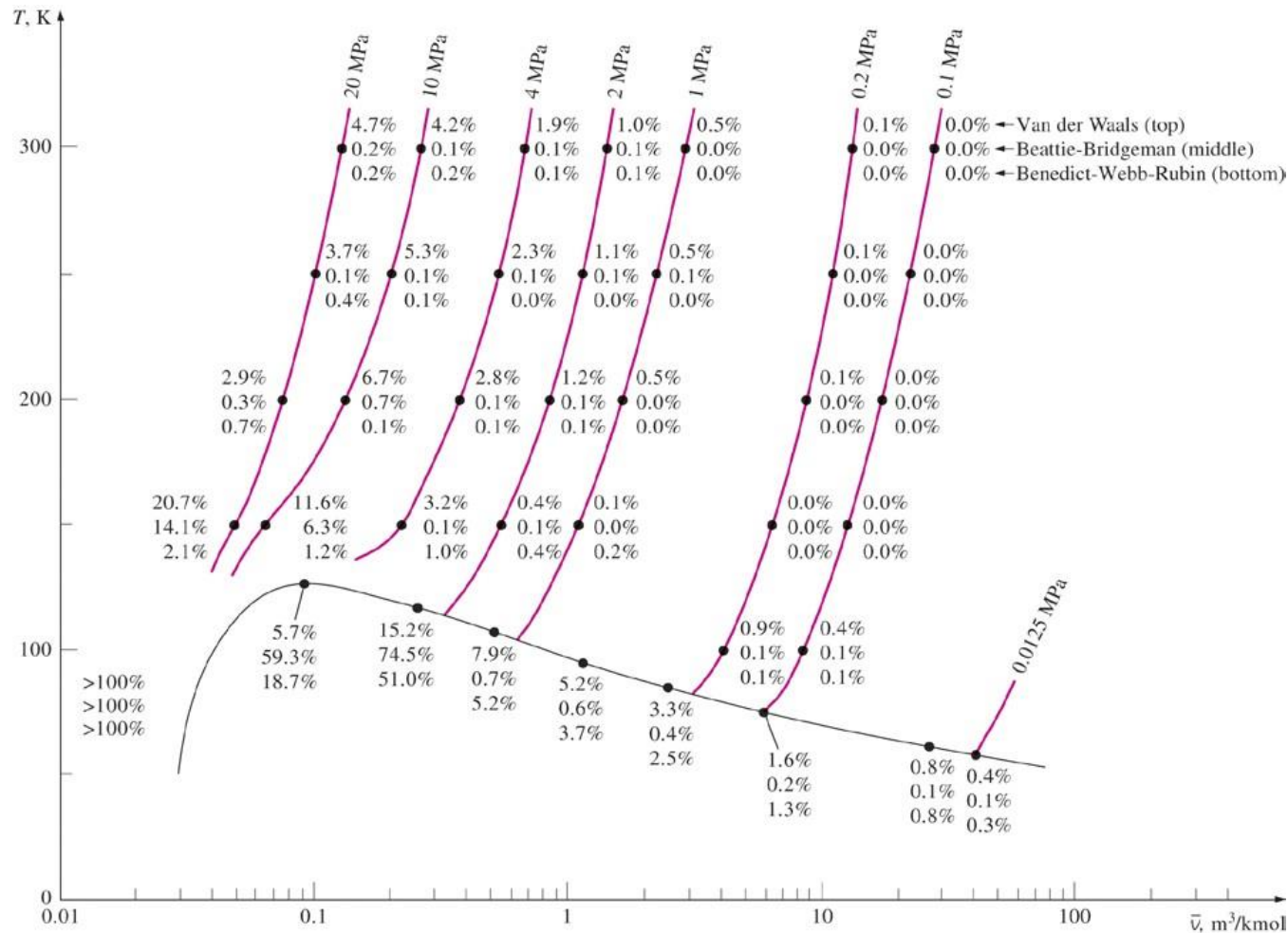
$$B = B_0 \left(1 - \frac{b}{\bar{V}} \right)$$

The constants are given in Table 3–4 for various substances. It is known to be reasonably accurate for densities up to about $0.8\rho_{cr}$.

Benedict-Webb-Rubin Equation of State

$$P = \frac{R_u T}{\bar{V}} + \left(B_0 R_u T - A_0 - \frac{C_0}{T^2} \right) \frac{1}{\bar{V}^2} + \frac{b R_u T - a}{\bar{V}^3} + \frac{a \alpha}{\bar{V}^6} + \frac{c}{\bar{V}^3 T^2} \left(1 + \frac{\gamma}{\bar{V}^2} \right) e^{-\gamma/\bar{V}^2}$$

The constants are given in Table 3–4. This equation can handle substances at densities up to about $2.5 \rho_{cr}$.



van der Waals: 2 constants.
Accurate over a limited range.

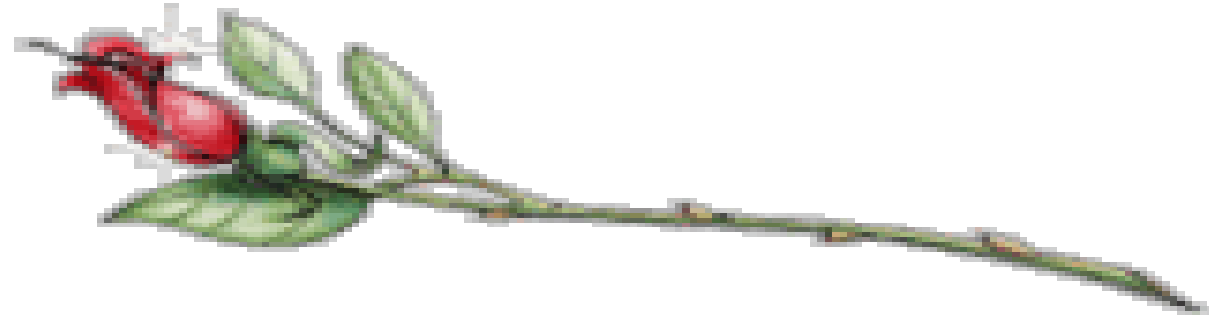
Beattie-Bridgeman: 5 constants.
Accurate for $\rho \leq 0.8\rho_{\text{cr}}$.

Benedict-Webb-Rubin: 8 constants.
Accurate for $\rho \leq 2.5\rho_{\text{cr}}$.

Strobridge: 16 constants.
More suitable for
computer calculations.

Virial: may vary.
Accuracy depends on the
number of terms used.

Complex equations of state represent the P - v - T behavior of gases more accurately over a wider range.



THANK YOU..😊